Multi-Hop Retrieval-Augmented Reasoning: A Latent Evidence-Chain Inference Perspective

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Multi-hop retrieval-augmented reasoning has been surveyed by architectural families such as graph reasoning, iterative retrieval, question decomposition, fusion-in-decoder reading, chain-of-thought prompting, and agentic search. These taxonomies are useful for cataloguing systems, but they hide a more basic statistical structure. The evidence chain required to answer a multi-hop question is *latent*, while the system only observes a noisy retrieval pool, a budgeted subset of that pool, an ordering of the selected evidence, and a generated answer.

This survey recasts multi-hop RAG as *latent evidence-chain inference* under missingness, noise, budget, exposure, and faithfulness constraints. Under this view, common failures correspond to distinct statistical events. The support chain may be absent from the retrieved pool, dropped by budgeted selection, exposed too late for the reader, incorrectly fused across evidence units, or used only post hoc rather than causally.

We make four contributions. First, we introduce a latent chain formalism that separates the unobserved support chain $C^\star$ from observable retrieval pools L_N, selected contexts E_K, evidence orderings $\pi(E_K)$, and generated answers $\hat{a}$. Second, we derive a success-factor decomposition that separates chain observability, conditional evidence utility, exposure probability, fusion reliability, and causal faithfulness, and provide minimax-style bounds explaining why no single module dominates. Third, we provide a comprehensive literature review organized along *two complementary axes*: an estimand axis that explains what statistical quantity a method targets, and an architectural axis that preserves continuity with the established literature. Across both axes we cover over 250 methods, benchmarks, and analyses, including recent 2024–2026 advances in agentic search, process-supervised retrieval, long-context reading, multimodal multi-hop reasoning, mechanistic interpretation, and latent reasoning. Fourth, we reinterpret benchmarks as diagnostic instruments and reformulate open problems as tradeoffs among recall, noise, context budget, reader exposure, causal grounding, and controllability, ending with a reporting standard that makes evidence-chain claims auditable across architectures.

CCS Concepts: • **Computing methodologies → Natural language generation**; **Information extraction**; • **Information systems → Question answering**.

Additional Key Words and Phrases: Multi-hop reasoning, retrieval-augmented generation, latent evidence chains, evidence retrieval, evidence fusion, faithfulness, causal attribution, large language models, knowledge graphs, agentic search, process supervision

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Foundations

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Foundations

1 Introduction

A multi-hop RAG system is rarely asked to retrieve a single fact. It is asked to recover an unobserved support structure. For a question q, the answer depends on a latent evidence chain $C^\star = (e_1^\star, \ldots, e_m^\star)$ whose elements must be discovered, ordered, and composed, while the system observes only a noisy retrieval pool L_N, a budgeted context E_K, an ordering $\pi(E_K)$, and the answer distribution induced by a reader. This mismatch between the latent object required for correctness and the observable objects available to the system is the central statistical difficulty of multi-hop retrieval-augmented reasoning.

Consider a question whose answer requires resolving an unstated bridge entity before the answer-bearing evidence can be found. A single-shot retriever sees only the surface form of the original question; it may retrieve highly relevant first-hop documents while missing the bridge document needed for the second hop. Increasing retrieval depth can make the complete chain observable, but it also increases distractor mass. Selecting a small context reduces noise, but it may discard a low-marginal-relevance bridge. Reordering the selected evidence can make the chain easier for the reader to use, but ordering cannot recover evidence that was never selected. Finally, even when the chain is present and visible, a fluent LLM may answer from parametric memory or produce a plausible chain that is not causally responsible for the answer. These are not independent engineering annoyances; they are successive statistical bottlenecks in estimating and using a latent support chain.

Existing surveys usually organize the field by architectural communities: graph-based reading comprehension, iterative retrieval, question decomposition, fusion-in-decoder readers, chain-of-thought prompting, and agentic search [235; 236; 242**? ?** −244]. This organization is useful for locating papers, but it is less useful for explaining why systems fail. Two methods with different surface architecture may target the same statistical bottleneck; conversely, two methods in the same family may fail for different statistical reasons. An iterative retriever, a graph expander, and an agentic search policy all attempt to increase the probability that the latent chain is observable. A reranker, a bridge selector, and a compression model all estimate some version of conditional evidence utility. Verification and counterfactual attribution, although often discussed separately, target causal faithfulness of the final answer.

This survey therefore asks a different organizing question: *which quantity of the latent evidence-chain inference problem does a method estimate?* The goal is not to discard architectural taxonomies, but to place them under a more diagnostic statistical layer. We use five estimands as the main coordinates: chain observability, conditional evidence utility, exposure probability, fusion reliability, and causal faithfulness. These estimands convert common multi-hop failure modes into measurable events and make design tradeoffs explicit. To preserve continuity with the existing literature, we additionally provide a parallel architectural review (Section 6) that catalogues methods by their surface design and maps each family to the estimands it primarily affects.

1.1 A Running Example through the Latent-Chain Lens

To make the distinction concrete, consider a question of the form:

> *"Which scientist discovered the chemical element that was named after the country where the author of* Cien años de soledad *was born?"*

A surface retriever may find documents about the novel and its author, but the latent chain contains at least four dependent units: $e_1^\star$, a biographical unit identifying that Gabriel García Márquez was born in Colombia; $e_2^\star$, a chemistry unit naming the element "columbium" (an older name for niobium) after Colombia or the related historical mineral; $e_3^\star$, a chemical history unit clarifying the identity of the element under its modern naming; and $e_4^\star$, an answer-bearing unit identifying the scientist (Charles Hatchett) who discovered the element. Three of the four units are not directly visible from the original question; they become queryable only after intermediate variables have been resolved.

We use this example, with minor variants, throughout the survey. In Section 3 we use it to ground the formalism $(C^\star, L_N, E_K, \pi(E_K), \hat{a})$; in Section 4 we use it to illustrate each of the five estimands; in Sections 5 and 6 we use it to show how different method families intervene at different points; in Section 7 we use it to discuss which benchmarks would expose which failure events; and in Section 9 we use it for counterfactual faithfulness illustrations.

In the notation used in this survey, the complete support chain $C^\star$ is not a document list returned by a system; it is the unobserved object that would make the answer identifiable. The retrieved pool L_N is an imperfect observation of that object, the selected context E_K is a budgeted estimator, and the reader-facing ordering $\pi(E_K)$ is a presentation of the estimator. This separation matters because different failures produce the same final symptom. If the answer is wrong, it may be because the birthplace document was absent from L_N, because the element document was dropped when E_K was selected, because the two documents were placed far apart in $\pi(E_K)$, because the reader failed to compose them, or because the model answered from memorized knowledge without using the chain. Final answer accuracy alone cannot distinguish these cases.

This example also explains why the statistical view is not merely a renaming of a pipeline. The latent chain induces a dependency structure, whereas the observed documents are sampled, selected, and ordered under system-specific policies. A retrieval module, a graph module, and an agentic search policy can therefore be compared by the same question: how much do they increase the probability that $C^\star$ becomes observable and usable under a fixed budget?

1.2 Position Relative to Existing Surveys

Prior surveys have studied general RAG [235; 236; 238; 240; 249], multi-hop QA [242; 243], complex KBQA [237; 239], knowledge-graph reasoning with LLMs [119; 124], chain-of-thought and step-by-step reasoning [234; 244; 245], hallucination [**? ?**], faithfulness and attribution [34; 85], and agentic RAG [246**? ?**]. A particularly close recent work is the four-axis framework for retrieval–reasoning processes in multi-hop QA [62], which organizes systems by execution plan, index structure, next-step control, and stopping criterion.

These surveys are complementary to ours but answer a different question. Their primary axes are typically retrieval mechanism, system architecture, reasoning style, data source, control flow, or application domain. The unit of analysis is procedural: *what does the system do at each step?* We add an estimand-based axis whose unit of analysis is statistical: *which latent quantity does the step attempt to estimate or improve?* Two procedures can have different executions but target the same estimand, and one procedure can target several estimands at once. For example, an agentic loop may simultaneously increase pool observability, estimate conditional evidence utility, and improve faithfulness through critique. Conversely, two iterative retrieval systems may both use repeated search but differ sharply in whether they optimize bridge observability or process faithfulness. The two views support different reviewer questions: a procedural view asks whether a method has a clear plan, index, trigger, and stopping rule; a latent-chain view asks whether the method improves $P(C^\star \subseteq L_N)$, preserves the chain under budget, exposes it early, composes it reliably, or uses it causally. We argue that a mature multi-hop RAG evaluation should report both, and this survey is structured so that the two views appear side by side.

The individual ingredients of our view are not new in isolation: retrieved documents have been treated as latent variables in end-to-end RAG [48; 82]; multi-hop QA has long used evidence and reasoning chains [55; 164; 204]; and recent attribution work connects faithfulness to causal evidence use [96; 190; 202]. The contribution of this survey is to connect these ingredients into a single statistical organization of multi-hop RAG, and to use that organization to support a comprehensive literature review whose method descriptions remain familiar to readers used to architectural categories.

1.3 Contributions

This survey makes four contributions.

- **Latent evidence-chain inference.** We recast multi-hop RAG as inference over an unobserved support chain $C^{\star}$. Retrieved pools, selected contexts, evidence orderings, and generated answers are treated as noisy observations or estimators of different aspects of that chain. We formalize the model as a latent-variable generative process and provide an MDP view that subsumes recent agentic and adaptive systems.
- **An estimand-based decomposition with theoretical guarantees.** We introduce five statistical estimands—chain observability, conditional evidence utility, exposure probability, fusion reliability, and causal faithfulness—and derive a success-factor decomposition. We accompany the decomposition with a minimax-style lower bound that explains why improving any single module yields bounded end-to-end gain whenever another module is weak, and with identifiability conditions linking observable diagnostics to latent events.
- **A comprehensive dual-axis review.** We review over 250 methods, benchmarks, and analyses along two complementary axes: an estimand axis that explains what statistical quantity a method targets, and an architectural axis (Section 6) that catalogues methods by graph reasoning, retrieval pipelines, question decomposition, fusion readers, LLM reasoning, agentic search, and hybrid LLM-with-KG approaches. We extend coverage to 2024–2026 advances including process-supervised retrieval, RL-based search agents, long-context reading, multimodal multi-hop reasoning, mechanistic interpretation, latent reasoning, and retrieval-side safety.
- **Statistical reinterpretation of benchmarks and open problems.** We reinterpret benchmarks as diagnostic instruments that make different latent events observable, propose reporting standards that align metrics with estimands, and reformulate common open problems as tradeoffs among missingness, noise, budget, exposure, fusion error, and causal grounding.

Scope. We focus on retrieval-augmented multi-hop reasoning over textual, structured, semi-structured, and hybrid knowledge sources. Single-hop QA, purely parametric reasoning without retrieval, general-purpose agents without retrieval-grounded evidence use, and multimodal multi-hop reasoning are discussed when they clarify the latent-chain view. Mechanistic interpretation and latent reasoning are included in dedicated subsections because they speak directly to whether the chain is even an explicit object in modern LLM-based pipelines.

Organization. The survey is organized into three parts. *Part I (Foundations)* contains the introduction (Section 1), survey methodology (Section 2), the latent-chain problem setup (Section 3), and the statistical decomposition (Section 4). *Part II (Literature Review)* contains the estimand-axis method review (Section 5), the architectural-axis review (Section 6), benchmarks (Section 7), and applications (Section 8). *Part III (Analysis and Outlook)* contains the faithfulness analysis (Section 9), open problems and tradeoffs (Section 10), reporting guidelines (Section 11), and conclusion (Section 12). Appendices supply method-level catalogues, mathematical notes, screening protocol, and a writing checklist.

2 Survey Methodology and Literature Coverage

A survey that proposes a new statistical organization must make its own selection procedure explicit. We screened papers along two dimensions: (i) whether the work directly addresses multi-step evidence dependency, and (ii) which latent-chain bottleneck it primarily targets. The first criterion prevents the survey from becoming a generic RAG or generic LLM-reasoning survey; the second prevents the taxonomy from collapsing into a list of architectural names.

2.1 Search and Inclusion Protocol

The literature pool was collected from ACL Anthology, ACM Digital Library, IEEE Xplore, arXiv, DBLP, Semantic Scholar, and Google Scholar, with particular attention to work published from 2017 onward. We used keyword families covering multi-hop question answering, complex question answering, retrieval-augmented generation, iterative retrieval, evidence retrieval, graph reasoning, KGQA, chain-of-thought, agentic search, process supervision, hallucination repair, citation faithfulness, counterfactual evaluation, long-context reading, and mechanistic interpretation of multi-hop reasoning. A paper was included when it satisfied at least one of the following conditions: it proposes a multi-hop method; it introduces a benchmark that exposes multi-step evidence dependency; it analyzes faithfulness or causal support in a retrieval-grounded setting; it is a foundational method that later multi-hop RAG systems reuse as a component; or it provides interpretability evidence about how multi-hop reasoning is implemented inside neural models.

We deliberately exclude papers that are only tangentially related. A single-hop retriever is discussed when it forms the upstream pool for a multi-hop system, but not as a full survey object. A general agent paper is discussed only if tool use is used to improve evidence acquisition, reasoning, or verification. A KG embedding method is discussed only when it functions as part of retrieval, graph expansion, or evidence fusion for multi-hop answering. This scope keeps the survey centered on latent evidence-chain inference rather than on every method that can be placed near RAG.

After title-and-abstract screening, candidate papers were assigned a primary architectural family (one of: retrieval, graph/KG, decomposition, fusion reader, LLM reasoning, agentic, hybrid, benchmark, analysis) and a primary estimand (one of: observability, utility, exposure, fusion, faithfulness, or joint). When a paper targeted multiple estimands, the dominant one was marked as primary and the others as secondary. Two independent passes were used on a sample of borderline cases to assess inter-screener agreement. Final selection prioritized methodological distinctiveness, citation impact, and contribution to estimand-level diagnosis.

2.2 Coverage Balance

The final taxonomy is intentionally balanced across historical and LLM-era work. Early systems such as DrQA [8], GRAFT-Net [151], PullNet [152], MDR [195], Baleen [74], FiD [57], REALM [48], DPR [72], and ColBERT [73; 136] establish the retrieval, graph, and fusion primitives. Middle-period systems such as IRCoT [165], Self-RAG [2], FLARE [64], Adaptive-RAG [60], and RAPTOR [137] make retrieval control and context organization explicit. Recent work from 2025–2026—including Search-R1 [71], DeepRAG [43], RAG-Gym [197], PAR-RAG [224], DualRAG [17], VERITAS [202], R1-Searcher [149], MultiCube-RAG [146]—moves the field toward learned search policies, plan-driven retrieval, process supervision, and faithfulness-aware reward design. We therefore treat LLM-era methods as new instantiations of the same evidence-chain problem, not as a clean break from earlier multi-hop QA.

In parallel, we deliberately include three threads that recent surveys underweight. The first is *long-context reading and retrieval* [4; 68; 98; 137; 214], which changes the granularity of evidence units. The second is *multimodal and table-text multi-hop reasoning* [7; 9; 11; 156; 159; 183], which

Table 1. Literature coverage organized by the primary object optimized. Counts are intended as a coverage audit rather than a claim of complete exhaustiveness.

Family	Approx. coverage	Core period	Representative role in the survey
Decomposition	20–25	2019–2026	Converts a complex query into lower-variance subqueries or executable steps.
CoT / search reasoning	25–30	2022–2026	Supplies reasoning topologies and search policies for chain construction.
Verification & faithfulness	15–20	2023–2026	Measures or improves step-level support and causal faithfulness.
Retrieval-based RAG	35–40	2017–2026	Controls chain observability and evidence utility under budget.
Agentic / RL search	25–30	2022–2026	Learns when, what, and how to retrieve in an interactive environment.
KG/ graph reasoning	25–30	2018–2026	Represents chains as paths or subgraphs and constrains fusion.
LLM +KG/ GraphRAG	20–25	2023–2026	Uses graph structure as a skeleton for LLM reasoning and generation.
Long-context / hierarchical	12–15	2023–2026	Reduces the number of explicit hops by changing retrieval granularity.
Multimodal / tabular	12–15	2020–2026	Broadens evidence substrates to images, tables, and structured records.
Mechanism / latent reasoning	8–10	2023–2026	Reveals or implies chain processes inside model internals.
Benchmarks and analyses	60+	2018–2026	Defines observable proxies for latent chain events.

broadens the substrate of $\mathcal{K}$ beyond passages. The third is *mechanistic interpretation and latent reasoning* [6; 38; 41; 51; 180; 208], which speaks to whether the chain is an explicit object inside the model or an implicit trajectory through its hidden states.

2.3 Positioning Against Recent Process-Centric Surveys

A particularly close recent survey is the four-axis framework for retrieval–reasoning processes in multi-hop QA [62], which organizes systems by execution plan, index structure, next-step control, and stopping criterion. That process view is complementary to ours. Its unit of analysis is the execution procedure: what the system does at each step. Our unit of analysis is the statistical estimand: which latent quantity the step attempts to estimate or improve. The two are orthogonal. A process feature such as "trigger retrieval when generation uncertainty exceeds a threshold" may serve different estimands depending on what the triggered retrieval optimizes (deeper bridge coverage, conditional utility, or faithfulness checking). Conversely, the same estimand may be improved by different procedural choices. We adopt the estimand view as the primary axis in Sections 4 and 5, but report architectural and procedural positions wherever they would help a reader trained in prior taxonomies.

Other complementary axes appear in surveys of LLM reasoning generally [234; 244; 245], of RAG architectures [235; 236; 238], and of agentic systems [241; 246**?**]. The estimand axis can be overlaid on each of these without contradiction.

3 Problem Setup: From Retrieved Documents to Latent Evidence Chains

3.1 What Makes a Question Multi-Hop?

A task is *multi-hop* when its solution requires connecting multiple evidence units whose discovery or use depends on intermediate states. Three properties characterize this dependency. First, *semantic non-locality*: at least one required evidence unit is not directly retrievable from the original question because its surface terms are absent. Second, *ordering sensitivity*: evidence must be combined through an ordered computation, such as bridge resolution, filtering, projection, comparison, or aggregation. Third, *joint sufficiency*: no strict subset of the support chain is enough to answer the question.

These properties distinguish multi-hop RAG from adjacent tasks. A long-document task may require high recall but remain single-hop if all relevant sentences are directly query-relevant. A multi-step generation task is not multi-hop unless different steps require different external evidence with cross-step dependency. A hard reasoning task with one piece of evidence is single-hop with difficult inference, not multi-hop retrieval. Returning to the running example, the question is multi-hop because the chemistry document about the element “columbium / niobium” is not retrievable from the surface terms of the literary question about *Cien años de soledad*: that document becomes queryable only after Colombia has been resolved as the bridge.

3.2 Latent Chain and Observable Variables

Let q be a question and $\mathcal{K} = \{e_1, e_2, \dots\}$ be a knowledge store. An evidence unit may be a passage, sentence, span, table cell, KG triple, or path. We denote the unobserved support chain required to answer q by

$$C^\star(q) = (e_1^\star, \dots, e_m^\star), \tag{1}$$

where the order encodes dependency among evidence units. The correct answer is $a^\star$.

A multi-hop RAG system does not observe $C^\star$ directly. It instead produces a sequence of observable objects:

$$\begin{aligned} L_N &= \mathcal{R}_N(q), && (2)\\ E_K &= S_K(q, L_N), \qquad E_K \subseteq L_N, \quad K \ll N, && (3)\\ \pi(E_K) &= \text{an ordering or presentation of } E_K, && (4)\\ \hat{a} &= \mathcal{G}(q, \pi(E_K)). && (5) \end{aligned}$$

Here L_N is the retrieved pool, E_K is the selected budgeted context, $\pi(E_K)$ is the reader-facing ordering, and $\hat{a}$ is the answer. This sequence defines the observable side of the latent-chain inference problem. For the running example, $C^\star$ has four ordered units (literary biography → Colombian historical-mineral trace → modern element identity → scientist discovery); L_N might contain biography passages, an unrelated chemistry overview, several distractors about Colombian geography, and possibly the answer-bearing passage; E_K is the top-K subset that the selector keeps; $\pi(E_K)$ determines whether the bridge passage is placed before or after the answer-bearing passage.

3.3 Failure Modes as Latent-Event Failures

The usual failure modes of multi-hop RAG can be written as failures of latent events:

Evidence absence.
$C^\star \not\subseteq L_N$. A required unit is missing from the retrieved pool, often at the bridge hop.

Budgeted selection failure.
$C^\star \subseteq L_N$ but $C^\star \not\subseteq E_K$. The chain is observable in the pool but is not preserved under a small context budget.

Table 2. Multi-hop RAG and adjacent research problems under the latent-chain view.

Adjacent problem	Shared aspect	Distinguishing aspect of multi-hop RAG
Single-hop open-domain QA	Retrieval over an external corpus	The latent object is a chain, not one directly retrievable fact.
Chain-of-thought reasoning	Multi-step intermediate text	Intermediate steps must be grounded in retrievable evidence, not only model-internal computation.
Compositional generalization	Combining components	Composition occurs over distributed evidence units, not only over latent functions.
KG path reasoning	Sequential relation traversal	A KG path is one possible chain substrate; text, tables, and hybrid stores also instantiate $\mathcal{K}$.
General hallucination detection	Detecting unsupported claims	Faithfulness must hold at every chain step, not only at the final claim.
Agentic LLM systems	Tool use and planning	Actions are evaluated by whether they improve latent-chain inference.
Long-context reading	Conditioning on long input	Granularity of evidence units shifts, but the chain may still be non-local and bridge-dependent.
Multi-document summarization	Combining many documents	Output is open-ended summary rather than a chain-identifiable answer.

Exposure failure.
$C^\star \subseteq E_K$ but the chain is placed too late or too diffusely in $\pi(E_K)$ for the reader to use reliably. This includes lost-in-the-middle [98] and ordering-gap effects.

Fusion or reasoning failure.
The chain is selected and visible, but the reader fails to compose the evidence units into the answer.

Faithfulness failure.
The answer is correct or plausible but is not causally supported by the retrieved chain. This includes incorrect citations and parametric leakage.

This event view is stricter than a natural-language taxonomy: it asks where, in the observable sequence $L_N \to E_K \to \pi(E_K) \to \hat{a}$, the latent support chain becomes unavailable, unusable, or non-causal. In the running example, "evidence absence" might be a missing chemistry passage; "budgeted selection failure" might be the bridge passage about Colombia being ranked below high-surface but unrelated documents; "exposure failure" might be the answer-bearing passage placed in position eight of an eight-passage prompt; "fusion failure" might be that the reader connects the wrong scientist to the element; and "faithfulness failure" might be that the reader answers correctly because the scientist is mentioned in pretraining data, while the cited chain does not actually entail the answer.

3.4 Adjacent Problems

3.5 Reasoning Types as Chain Structures

The latent-chain view gives a compact way to describe common multi-hop reasoning types. *Bridge reasoning* corresponds to a chain in which an early evidence unit resolves an unmentioned entity needed to retrieve a later unit. *Comparison reasoning* corresponds to two or more parallel subchains whose terminal attributes must be compared. *Compositional reasoning* corresponds to a sequence of operators such as filter, project, aggregate, and compare applied over distributed evidence [158; 164]. *Verification-oriented reasoning* corresponds to a chain whose terminal object is not an answer string but a support or refutation label [63; 101; 162]. *Counterfactual probing* replaces one node or relation in $C^\star$ and tests whether the induced answer distribution changes [190].

These reasoning types differ in the shape of $C^\star$. A bridge question is often close to a path; a comparison question is closer to a small tree; a fact-verification problem may require a set of jointly supporting facts; and a table-text question may alternate between cells, linked passages, and entity mentions [9; 11]. The statistical bottlenecks remain the same. The system must make the relevant structure observable, select it under budget, expose it to the reader, compose it correctly, and use it causally. This is why the same estimand-based vocabulary can cover graph paths, text passages, tables, and agent trajectories.

3.6 A More Explicit Generative View

The preceding notation can be written as a latent-variable model. Let $\mathfrak{C}(q)$ denote the set of admissible evidence chains for q. A dataset or world distribution first samples a valid chain $C \in \mathfrak{C}(q)$, then an answer is generated from the chain:

$$C \sim p(C \mid q, \mathcal{K}), \tag{6}$$
$$a^\star \sim p(a \mid q, C). \tag{7}$$

A retrieval-augmented system observes only a corrupted view of this chain through retrieval and context construction:

$$L_N \sim p_R(L \mid q, \mathcal{K}), \tag{8}$$
$$E_K \sim p_S(E \mid q, L_N, K), \tag{9}$$
$$\pi \sim p_O(\pi \mid q, E_K), \tag{10}$$
$$\hat{a} \sim p_G(a \mid q, \pi(E_K)). \tag{11}$$

Thus the ideal Bayesian objective is

$$\hat{a}_{\text{Bayes}} = \arg\max_a \sum_{C \in \mathfrak{C}(q)} p(a \mid q, C) p(C \mid q, \mathcal{P}_{\text{sys}}), \tag{12}$$

where $\mathcal{P}_{\text{sys}} = (L_N, E_K, \pi(E_K))$ are the observations made available by the system. Real systems approximate this posterior through retrieval scores, rerankers, graph traversals, prompts, tool calls, and verifiers. The statistical question is how each approximation changes the posterior mass assigned to complete and causally sufficient chains.

Definition 1 (Minimal support chain)**.** For a question q and answer $a^\star$, a chain $C = (e_1, \ldots, e_m)$ is *sufficient* if $p(a^\star \mid q, C)$ exceeds a task-specific correctness threshold. It is *minimal* if no proper subsequence of C is sufficient. The set of all minimal chains is denoted by $\mathfrak{C}^\star(q)$.

This definition is deliberately set-valued. Many questions admit multiple valid chains. A Wikipedia biography, a table row, and a KG path may all support the same answer. Treating $C^\star$ as a single sequence is therefore a notational convenience; the more faithful object is a set or distribution over minimal chains. When the survey writes $C^\star \subseteq L_N$, the statement should be read as "there exists at least one valid minimal chain fully contained in the pool."

Definition 2 (Multi-hop dependency graph)**.** Given a minimal chain $C = (e_1, \ldots, e_m)$, define a directed graph $D_C = (V, A)$ whose vertices are evidence units and whose edge $e_i \longrightarrow e_j$ indicates that e_j becomes identifiable, retrievable, or interpretable only after information in e_i is known. The hop count is the length of the longest directed path in D_C.

This graph definition clarifies why hop count is not identical to the number of retrieved passages. A comparison question may have two parallel single-hop branches but still be multi-hop because the final comparison operator depends on both branches. A bridge question may have a path-like structure. A table-text question may alternate between table cells and linked passages. The chain formalism abstracts over these substrates.

Assumption 1 (Budgeted observability)**.** For a fixed final context budget K, a multi-hop system can answer only if at least one minimal chain has cardinality at most K after the chosen evidence-unit granularity is fixed.

Assumption 1 is not a statement about the world; it is a statement about an evaluation protocol. If the minimal chain requires six passages but the reader receives only five, then no selector can preserve a passage-level chain. The only possible remedies are to change the granularity, compress evidence, use a larger budget, or perform iterative generation with additional retrieval calls.

3.7 State-Space and MDP Formulations

Many recent agentic and adaptive RAG methods can be written as a controlled stochastic process [2; 43; 60; 71; 197]. Let $s_t = (q, h_t, L_t, M_t)$ be the state at step t, where h_t is the reasoning history, L_t is the accumulated retrieval pool, and M_t is a memory or scratchpad. An action u_t may be a query rewrite, retrieval call, graph expansion, compression step, verification call, or final-answer action. The transition is

$$s_{t+1} \sim P(s_{t+1} \mid s_t, u_t, \mathcal{K}), \tag{13}$$

and the terminal reward can combine correctness, faithfulness, and cost:

$$R_T = \mathbb{I}\{\hat{a} = a^\star\} - \lambda_c \sum_{t=0}^{T-1} \mathrm{cost}(u_t) - \lambda_f \mathbb{I}\{\hat{a} \not\leftarrow C^\star\}. \tag{14}$$

This casts agentic RAG as policy learning:

$$\pi^\star = \arg\max_{\pi} \mathbb{E}_{\pi,P}\left[R_T\right]. \tag{15}$$

DeepRAG [43], Search-R1 [71], R1-Searcher [149], ReSearch [15], and RAG-Gym [197] differ mainly in what state variables are observable, how actions are parameterized, whether rewards are outcome-only or process-supervised, and how retrieved tokens are handled during optimization.

The MDP view is useful but incomplete. The true support chain is still latent; the policy receives sparse and biased observations. Therefore, agentic search does not remove the latent-chain problem. It merely moves the approximation from a fixed retriever to a learned control policy over retrieval and reasoning actions.

3.8 Where Does the Chain Live? Latent Reasoning and Mechanistic Interpretation

A separate but related thread asks whether the chain is even an explicit object inside modern LLM-based systems. *Latent reasoning* methods compress some or all reasoning steps into hidden states rather than tokens. Quiet-STaR [41] trains models to silently generate internal rationales before each token. Coconut [51] performs continuous chain-of-thought in a latent space, decoupling reasoning from natural language. Pause tokens and "think before speak" variants insert non-emitted computation [40]. From the latent-chain view, these methods do not eliminate $C^\star$; they make it unobservable through tokens, which complicates both diagnostic evaluation and faithfulness attribution.

Mechanistic interpretation of multi-hop reasoning takes the opposite stance: it locates traces of $C^\star$ inside model internals. Probing work has shown that LLMs sometimes recall bridge entities in intermediate layers [6; 38; 208], that the recall of a second-hop fact can be circuit-localized [180], and that some questions are answered through "shortcut" rather than genuine bridge recall [208]. These findings matter for this survey for two reasons. First, they suggest that even when no chain is written into the prompt, an implicit chain can sometimes be recovered from internal activations. Second, they motivate diagnostics that combine retrieved-evidence sensitivity with internal-state sensitivity: a model whose answer changes when an internal bridge representation is patched is more plausibly using the chain causally than one whose answer is invariant under such interventions.

We return to this dimension in Section 9 when discussing faithfulness diagnostics, and in Section 10 when discussing open problems for latent reasoning under retrieval augmentation.

4 A Statistical Decomposition of Multi-Hop RAG

This section introduces the five statistical quantities used to organize the rest of the survey. Each quantity corresponds to one stage at which a latent support chain can be lost or misused.

4.1 Chain Observability

The first question is whether the complete support chain appears in the retrieved pool:

$$\theta_{\text{obs}}(q, N) = P(C^{\star} \subseteq L_N \mid q). \tag{16}$$

This quantity captures retrieval-side missingness. If θ_{obs} is low, no selector, reader, or verifier can recover a reliable chain from the pool. When a chain has m hops and each required unit has marginal retrieval probability p_i, then under an independence approximation,

$$P(C^{\star} \subseteq L_N) \approx \prod_{i=1}^{m} p_i. \tag{17}$$

This explains a basic scaling law of multi-hop difficulty: even modest per-hop miss rates compound rapidly as the chain length grows. Empirically, on MuSiQue [164] 4-hop subsets, many single-shot retrievers achieve high recall on the first hop but nearly zero pool-level chain recall, exactly as the product form predicts.

For the running example, θ_{obs} depends on whether the biography passage, the Colombia-to-element bridge passage, the element-identity passage, and the scientist-discovery passage all appear in L_N. A single-shot dense retriever conditioned only on the literary surface query is unlikely to retrieve the third and fourth units; the conditional probability of finding them rises only after "Colombia" has been resolved.

4.2 Conditional Evidence Utility

Once the chain is observable in L_N, the system must select a budgeted subset E_K. Standard retrievers estimate marginal relevance $s_{\text{rel}}(q, e)$, but multi-hop selection requires conditional utility:

$$\theta_{\text{util}}(e \mid q, C) = P(a^{\star} \mid q, C \cup \{e\}) - P(a^{\star} \mid q, C). \tag{18}$$

The key statistical mismatch in multi-hop retrieval is that a document with low marginal relevance may have high conditional utility once a partial chain C is known. Bridge documents often look weak under query-only scoring but are decisive under core-conditioned scoring. Reranking, bridge selection, evidence compression, and core-conditioned selection all target variants of this quantity [74; 199; 216; 224].

4.3 Exposure Probability

Selection is not enough. A reader with limited attention or positional bias may fail to use a complete chain if the relevant units are late, scattered, or separated by distractors. We define exposure probability as

$$\theta_{\text{exp}}(q, K, r) = P\big(\text{rank}_\pi(C^\star) \le r \mid C^\star \subseteq E_K\big), \tag{19}$$

where $\text{rank}_\pi(C^\star)$ denotes the latest or most difficult-to-access position of the required chain under ordering π, and r is a reader-dependent "effective access depth" beyond which positional bias starts to dominate [98]. Ordering, reader-aware reranking, context packing, and compression methods target this quantity.

4.4 Fusion Reliability

When the chain is present and exposed, the reader must compose it. We define fusion reliability as

$$\theta_{\text{fus}}(q) = P(\hat{a} = a^\star \mid C^\star \subseteq E_K, \text{rank}_\pi(C^\star) \le r). \tag{20}$$

This quantity separates retrieval success from reasoning success. It explains why high evidence recall does not imply high answer accuracy: the reader may fail to compare, aggregate, bridge, or verify the selected evidence.

4.5 Causal Faithfulness

Finally, correctness does not imply faithful evidence use. We write causal faithfulness as

$$\theta_{\text{faith}}(q) = P(\hat{a} \leftarrow C^\star \mid \hat{a} = a^\star), \tag{21}$$

where $\hat{a} \leftarrow C^\star$ means that the answer is causally supported by the chain rather than merely co-occurring with it or being reconstructed from parametric memory. The corresponding faithfulness gap is

$$\Delta_{\text{faith}} = P(\hat{a} = a^\star,\ \hat{a} \not\leftarrow C^\star). \tag{22}$$

Verification, citation checking, counterfactual evaluation, and process-supervised reward models all attempt to reduce this gap [37; 39; 94; 155; 190; 202; 203; 226].

4.6 Success-Factor Decomposition

Let

$$\mathcal{P} = \{C^\star \subseteq L_N\}, \tag{23}$$

$$\mathcal{S} = \{C^\star \subseteq E_K\}, \tag{24}$$

$$\mathcal{O} = \{\text{rank}_\pi(C^\star) \le r\}, \tag{25}$$

$$\mathcal{U} = \{\mathcal{G} \text{ correctly composes } C^\star\}, \tag{26}$$

$$\mathcal{T} = \{\hat{a} \text{ is causally supported by } C^\star\}. \tag{27}$$

Then final answer success can be approximated as a product of conditional bottlenecks:

$$P(\hat{a} = a^\star) \approx P(\mathcal{P})P(\mathcal{S} \mid \mathcal{P})P(\mathcal{O} \mid \mathcal{S})P(\mathcal{U} \mid \mathcal{O})P(\mathcal{T} \mid \mathcal{U}). \tag{28}$$

Remark 1 (On the meaning of equation 28). This expression is presented as a *diagnostic factorization*, not as a claim of conditional independence. The right-hand side is an exact chain-rule decomposition over a particular nested-event sequence $\mathcal{P} \Rightarrow \mathcal{S} \Rightarrow \mathcal{O} \Rightarrow \mathcal{U} \Rightarrow \mathcal{T}$; the approximation symbol acknowledges that real systems include parametric leakage and shortcut events that allow $\hat{a} = a^\star$ even when some upstream event fails. The decomposition is useful even when the events are dependent because (i) it identifies five distinct bottlenecks each of which can be diagnosed on

Table 3. Estimand-based taxonomy of multi-hop RAG. Methods are grouped by the statistical quantity they primarily estimate or improve.

Estimand	Mathematical form	Diagnosed bottleneck	Representative mechanisms
Chain observability	$P(C^\star \subseteq L_N)$	Evidence missingness	Iterative retrieval, graph expansion, agentic search
Conditional utility	$P(a^\star \mid C \cup \{e\}) - P(a^\star \mid C)$	Budgeted selection bias	Reranking, bridge selection, compression, core-conditioned selection
Exposure probability	$P(\text{rank}_\pi(C^\star) \le r)$	Ordering and lost-in-the-middle	Reader-aware ordering, passage packing, context compression
Fusion reliability	$P(\hat{a} = a^\star \mid C^\star \subseteq E_K)$	Cross-evidence reasoning failure	FiD, graph fusion, decomposition, programmatic reasoning
Causal faithfulness	$P(\hat{a} \leftarrow C^\star \mid \hat{a} = a^\star)$	Correct but unsupported answers	Verification, citation checking, counterfactual attribution, PRM

conditional subsets, and (ii) it provides an upper bound: $P(\hat{a} = a^\star)$ never exceeds the minimum of the five factors taken in isolation under the no-shortcut-leakage condition.

Theorem 1 (Minimax bottleneck bound)**.** *Assume the reader has no parametric leakage and that $\hat{a} = a^\star$ implies $\mathcal{T}$ holds (i.e., correct answers are causally grounded). Then for any system whose product factorization takes the form of equation 28,*

$$P(\hat{a} = a^\star) \;\le\; \min_{i \in \{\text{obs,sel,exp,fus,faith}\}} \theta_i, \tag{29}$$

where the right-hand side denotes the unconditional version of each estimand. Consequently, the marginal effect of improving estimand θ_i on end-to-end success satisfies

$$\frac{\partial P(\hat{a} = a^\star)}{\partial \theta_i} \;\le\; \prod_{j \ne i} \theta_j. \tag{30}$$

Proof sketch. The chain-rule decomposition implies each conditional factor is upper bounded by its unconditional counterpart, since conditioning on a larger event in a nested sequence can only redistribute mass. Under the no-leakage assumption, the joint success requires every event to hold, so $P(\hat{a} = a^\star)$ is dominated by the smallest factor. Differentiating the product expression yields the marginal bound. A fully rigorous treatment requires regularity conditions on the joint distribution of the five events and is given in Appendix B. □

Theorem 1 formalizes a common empirical pattern in multi-hop RAG: a strong retriever does not improve answer accuracy when the reader cannot fuse evidence; a strong reader does not help when the chain is absent from the pool; a faithfulness verifier may reject bad answers but cannot recover missing evidence. It also implies that the most credible empirical papers are those that show *where in the product* the gain occurs.

4.7 Risk under Missingness, Noise, and Budget

The estimand view also yields a compact risk perspective. Let ρ_N be the noise ratio of the retrieval pool,

$$\rho_N = \frac{|L_N \setminus C^\star|}{|L_N|}. \tag{31}$$

A stylized system risk can be written as

$$\mathcal{R}(N,K) = \alpha P(C^\star \nsubseteq L_N) + \beta\mathbb{E}[\rho_K] + \gamma\,\mathrm{Cost}(N,K) + \delta\Delta_{\mathrm{faith}}. \tag{32}$$

This expression makes explicit why multi-hop RAG is not solved by "retrieve more" or "use a larger reader." Deeper retrieval reduces missingness but increases noise and cost; smaller contexts reduce noise but increase selection bias; stronger readers may improve fusion while leaving faithfulness unresolved.

4.8 Identifiability and Surrogate Supervision

The five estimands are conceptually clean but not always directly observable. In most datasets, $C^\star$ is partially annotated, non-unique, or absent. Supporting-fact labels are only a surrogate for chain membership; they may omit alternative valid chains, include facts that are sufficient but not necessary, or differ from the path actually used by the model. This creates an identifiability problem: answer accuracy can be observed, but the latent event that failed is often not identified without additional instrumentation.

Proposition 1 (Conditional identifiability)**.** *Suppose for each example we observe a set $\Sigma(q) \subseteq \mathfrak{C}^\star(q)$ of annotated valid chains and the full context sequence $(L_N, E_K, \pi(E_K), \hat{a})$. Then the events $\mathcal{P}$, $\mathcal{S}$, and $\mathcal{O}$ are identifiable up to chain-set incompleteness: an observed positive $\mathbb{I}\{\Sigma(q) \subseteq L_N\}$ implies $\mathcal{P}$ holds, while an observed negative is informative only if $\Sigma(q)$ covers $\mathfrak{C}^\star(q)$. The events $\mathcal{U}$ and $\mathcal{T}$ are not identifiable from $(\Sigma(q), L_N, E_K, \pi(E_K), \hat{a})$ alone; they require interventional probes such as evidence deletion, bridge counterfactual, or oracle-evidence comparison.*

We distinguish three supervision regimes. In *fully annotated* regimes, path or step labels [19; 55; 164; 204] allow approximate measurement of observability and selection. In *weakly annotated* regimes, supporting facts or citations provide noisy labels for membership and faithfulness. In *unannotated* regimes, the system must rely on proxies such as retrieval overlap, counterfactual perturbation, leave-one-out deletion [96], or verifier scores [202].

These regimes should not be mixed without qualification. A method that improves answer F1 under unannotated evaluation may have improved observability, fusion, or parametric guessing; without estimand-level diagnostics the mechanism is underdetermined. This perspective suggests a reporting standard for future work (Section 11): for any multi-hop system, authors should report at least one metric for each of three stages: pool-level chain coverage, context-level chain preservation, and answer-level faithfulness.

4.9 A Diagnostic Protocol

The statistical decomposition induces a simple diagnostic protocol. First, evaluate the retrieved pool L_N against available support labels to estimate chain observability. If the chain is absent, downstream reader errors should not be attributed to reasoning. Second, condition on chain-present examples and measure whether the selected budgeted context E_K preserves the chain. This isolates selection bias under budget. Third, hold membership fixed and vary the ordering $\pi(E_K)$ to measure exposure effects. Fourth, evaluate the reader under oracle or near-oracle evidence to estimate fusion reliability. Fifth, perturb or remove cited evidence to test causal faithfulness.

The important point is that these diagnostics are conditional. A faithfulness test on examples where the evidence is absent conflates retrieval and attribution. A fusion test on examples where the chain is not selected conflates selection and reasoning. A ranking test that changes both membership and order cannot isolate exposure. Conditional evaluation is therefore not a minor experimental detail; it is required by the latent-chain factorization itself.

Table 4. Estimand-level diagnostics for multi-hop RAG. Each metric is best interpreted conditionally on the preceding event.

Stage	Event	Possible diagnostic	Common confounder
Pool construction	$\mathcal{P} = \{C^\star \subseteq L_N\}$	pool support recall, path recall, bridge recall	shortcut labels or incomplete gold chains
Budgeted selection	$\mathcal{S} = \{C^\star \subseteq E_K\}$	selected support recall, chain preservation rate	relevance-biased gold annotations
Presentation	$\mathcal{O} = \{\mathrm{rank}_\pi(C^\star) \le r\}$	early support exposure, support adjacency, lost-in-middle sensitivity	simultaneous membership changes
Fusion	$\mathcal{U}$	oracle-evidence EM/F1, step accuracy, path validity	parametric answer leakage
Faithfulness	$\mathcal{T}$	citation precision, deletion tests, counterfactual sensitivity	post-hoc rationalization

4.10 Monotonicity, Interaction, and Submodularity

A common implicit assumption in retrieval pipelines is that evidence utility is approximately additive: if each document is relevant on its own, the set should be useful. Multi-hop reasoning violates this assumption. Let $F_q(E)$ denote the probability that a fixed reader answers correctly given evidence set E:

$$F_q(E) = P(\hat{a} = a^\star \mid q, E). \tag{33}$$

The marginal utility of an evidence unit e under current context C is

$$\Delta_q(e \mid C) = F_q(C \cup \{e\}) - F_q(C). \tag{34}$$

A single-hop task is often close to monotone submodular selection: adding a relevant document helps and marginal gains diminish. Multi-hop tasks are frequently non-submodular because a bridge document may have near-zero utility alone but large utility after its companion evidence is present.

Definition 3 (Complementarity gap). For evidence units e_i, e_j and context C, define

$$\Gamma_q(e_i, e_j \mid C) = \Delta_q(e_j \mid C \cup \{e_i\}) - \Delta_q(e_j \mid C). \tag{35}$$

Positive Γ_q indicates complementarity: the value of e_j increases after e_i is known.

Proposition 2 (Why query-only relevance misses bridges). *Suppose a bridge evidence unit e_b has low marginal relevance $s_{\mathrm{rel}}(q, e_b)$ but positive complementarity $\Gamma_q(e_c, e_b \mid C) > 0$ with a core unit e_c. Any selector that is monotone in $s_{\mathrm{rel}}(q, e)$ alone can rank e_b below unrelated high-surface-match distractors even when $\Delta_q(e_b \mid C \cup \{e_c\})$ is high.*

Proof. A query-only selector is a function of $s_{\mathrm{rel}}(q, e)$ and does not condition on $C \cup \{e_c\}$. If there exists a distractor d such that $s_{\mathrm{rel}}(q, d) > s_{\mathrm{rel}}(q, e_b)$, the selector will prefer d regardless of the conditional gain of e_b. The inequality can hold even when $F_q(C \cup \{e_c, e_b\}) > F_q(C \cup \{e_c, d\})$, because that comparison depends on conditional utility rather than marginal surface relevance. The argument requires only that the selector's score function depends on (q, e) and not on C, and that the strict-inequality regime is non-empty—a condition that is empirically verified on bridge-heavy benchmarks such as MuSiQue [164] and 2WikiMultiHopQA [55]. □

This proposition is the formal reason why multi-hop reranking should not be evaluated only as relevance reranking. A selector may reduce average query-document relevance while improving chain preservation. Conversely, a strong cross-encoder reranker may improve top-K relevance while still dropping low-surface bridge evidence.

4.11 A Budgeted Chain-Preservation Objective

Given a pool L_N, context budget K, and cost function $c(e)$, an ideal selector solves

$$E_K^\star = \arg\max_{E \subseteq L_N} \mathbb{E}_{C \sim p(C|q,L_N)} \left[\mathbb{I}\{C \subseteq E\}\right] \quad \text{s.t.} \quad |E| \leq K. \tag{36}$$

If evidence units have variable length, the cardinality budget becomes a token budget:

$$\sum_{e \in E} c(e) \leq B. \tag{37}$$

Compression methods can be interpreted as changing $c(e)$ and sometimes changing the evidence unit itself. A compressed note may preserve the answer-bearing fact but lose the bridge relation or the provenance needed for faithfulness. Therefore compression should be evaluated not only by answer accuracy but by chain preservation and citation validity [199; 216].

Lemma 1 (Upper bound under chain absence). *Let A be the event that no valid minimal chain is contained in the retrieved pool L_N. If the reader is required to answer only from the retrieved evidence and has no parametric leakage, then*

$$P(\hat{a} = a^\star \mid A) \leq \epsilon, \tag{38}$$

where ϵ is the probability of guessing correctly from task priors or annotation artifacts.

Proof. Under the no-leakage condition, the answer is identifiable only from a sufficient evidence chain. When no valid minimal chain is present, the reader lacks the information required to distinguish $a^\star$ from alternatives except through prior bias. The remaining success probability is therefore bounded by shortcut or guessing probability ϵ. Formally, let $\Pi(a \mid A)$ be the conditional answer distribution when no chain is present. Then for any reader output $\hat{a}$, $P(\hat{a} = a^\star \mid A) = \Pi(a^\star \mid A) \leq \max_a \Pi(a \mid A) \leq \epsilon$ by the assumption on prior coverage. □

The lemma motivates conditional evaluation. If a method fails on examples where L_N lacks the chain, the failure should be attributed to observability, not to the reader. If a method succeeds in that regime, the success may indicate parametric memory or shortcut exploitation rather than retrieval-grounded reasoning.

4.12 Exposure as a Positional Treatment

Exposure can be treated as a causal treatment over a fixed evidence set. Let $Y(\pi)$ denote the answer-correctness outcome under ordering π. The average exposure effect between two ordering policies π_1 and π_0 is

$$\tau_{\text{exp}} = \mathbb{E}\left[Y(\pi_1) - Y(\pi_0) \mid C^\star \subseteq E_K\right]. \tag{39}$$

This estimand is identifiable only if membership is held fixed. If the reranker changes both E_K and $\pi(E_K)$, gains conflate selection and presentation. This is why fixed-membership ordering experiments are not secondary ablations; they are required to isolate the exposure channel, especially on long contexts where positional bias is documented to be substantial [4; 98; 214].

4.13 Reader Fusion as Noisy Program Execution

A multi-hop reader can be idealized as executing a latent program over evidence units:

$$z_0 = q, \qquad z_t = T_t(z_{t-1}, e_t), \qquad \hat{a} = H(z_m), \tag{40}$$

where T_t may resolve an entity, project an attribute, compare values, or aggregate partial results. If each transition has error probability ϵ_t, then under a simple independent-error approximation,

$$P(\text{all transitions correct}) \approx \prod_{t=1}^{m}(1 - \epsilon_t). \tag{41}$$

This explains why longer chains are difficult even when every document is retrieved. It also explains why process supervision [94; 155; 197], program execution [12; 36; 100], external tools [139; 209], and step-level verification can improve multi-hop RAG without changing retrieval recall.

Literature Review
Literature Review

5 Methods as Estimators of Evidence-Chain Quantities

This section reorganizes the method landscape by statistical estimand. Architectural families reappear, but as implementation choices rather than primary organizing categories. The subsections below deliberately cut across architecture. For example, graph expansion, iterative dense retrieval, and agentic search all appear under chain observability because they expand or adapt the set of candidate evidence units. Decomposition appears under fusion reliability when it reduces the complexity of composition, but it also appears under observability when sub-questions become retrieval queries. This overlap is not a flaw; it is the point of an estimand-based taxonomy. A method can be an estimator of more than one quantity, and a strong system often combines several estimators. Readers who prefer an architecture-first presentation should read this section together with Section 6, which catalogues the same methods under their conventional family labels.

5.1 Estimating Chain Observability

Methods in this group aim to increase $P(C^\star \subseteq L_N)$. We organize them along increasing degrees of adaptivity: single-shot retrieval, iterative retrieval, graph expansion, and agentic search.

5.1.1 Single-shot Sparse and Dense Retrieval. Sparse retrievers such as BM25 [135] and tuned BM25 variants remain strong baselines for multi-hop QA when at least one hop is lexically anchored. Dense retrievers including DPR [72], REALM [48], ANCE [194], Contriever [59], GTR [116], E5 [173], BGE [193], and RetroMAE [192] provide scalable proposal pools by encoding question and passage into a shared dense space. Multi-vector and late-interaction retrievers such as ColBERT [73] and ColBERTv2 [136] preserve per-token matching signals that are useful when bridge entities appear only in narrow spans. Learned sparse retrievers such as SPLADE [32] combine inverted-index efficiency with neural expansion. The shared limitation of single-shot retrievers in the multi-hop setting is structural: if the second-hop bridge is not expressed in the original query, query-only retrieval may miss it, regardless of whether scoring is sparse, dense, or multi-vector.

5.1.2 Iterative and Reasoning-Interleaved Retrieval. Iterative retrieval directly addresses second-hop missingness. GoldEn Retriever [112] generates intermediate queries from previously retrieved passages. MDR [195] performs multi-step dense retrieval conditioned on accumulated evidence representations. Baleen [74] maintains a condensed evidence state across hops to avoid query drift. MUPPET [28] pre-encodes potential bridge links and reuses them between hops. IRCoT [165] interleaves chain-of-thought reasoning with retrieval, treating each reasoning step as a new query.

FLARE [64] triggers retrieval mid-generation whenever predicted-token probability falls below a threshold. ITER-RETGEN [144] alternates between generation-as-context-expansion and retrieval-as-evidence-refresh. Adaptive-RAG [60] classifies query complexity to select between no-retrieval, single-step, and multi-step retrieval paths. DRAGIN [150] decides retrieval timing based on generator entropy and uncertainty signals. Self-RAG [2] uses learned reflection tokens to decide both when to retrieve and which passages to keep. ITER-DRAGON [191] composes iterative retrieval with stage-wise verifier feedback.

5.1.3 Graph Expansion as Observability Estimator. Graph-expansion methods can also be read as observability estimators. GRAFT-Net [151] and PullNet [152] construct question-specific subgraphs by iteratively pulling neighboring entities and linked text. HGN [27] and DFGN [133] build hierarchical entity-and-paragraph graphs to surface bridge nodes that single-shot retrieval misses. ToG [154], ToG 2.0 [103], and RoG [99] traverse KG paths under LLM control, expanding the local search space until candidate paths become visible. KG-FiD [215] uses KG signals to prune and re-score retrieved passages. GNN-RAG [109] runs a GNN over a retrieved subgraph and feeds the resulting node representations to an LLM reasoner. GraphRAG [25] constructs an LLM-built graph index over a corpus and uses community summaries for query-focused expansion. LightRAG [46] and HippoRAG [47] provide alternative graph indexing strategies optimized for cost and memory-style retrieval.

5.1.4 Long-context and Hierarchical Retrieval. A different way to increase chain observability is to change the granularity of evidence units. Long-context readers can sometimes absorb the entire candidate pool, but lost-in-the-middle effects [98] prevent naive scaling. RAPTOR [137] builds recursive abstractive summaries of corpus clusters so that high-level bridge information becomes retrievable. LongRAG [68] pairs long-context readers with coarser retrieval units to reduce per-hop selection load. HiQA [13] introduces hierarchical context augmentation. HyDE [35] synthesizes hypothetical documents to anchor dense retrieval in semantic rather than lexical space, and is especially useful for queries whose surface form does not match relevant evidence.

5.1.5 Agentic and RL-trained Search Policies. Recent work moves observability estimation into a learned policy. ReAct [209] interleaves reasoning and acting through tool calls including search. Toolformer [139] learns when to invoke external tools through self-supervision. Search-R1 [71] trains a reasoning model to generate search queries during step-by-step reasoning with reinforcement learning, with retrieval results masked out from policy gradients. R1-Searcher [149] incentivizes the search capability of LLMs through RL with intermediate retrieval rewards. Deep-RAG [43] formulates retrieval-augmented reasoning as an MDP and decides at each step whether to retrieve or rely on parametric reasoning. RAG-Gym [197] adds process supervision for search agents, making intermediate search actions trainable. PAR-RAG [224] uses a plan-act-review structure to reduce reasoning-path drift. DualRAG [17] couples reasoning-augmented querying with progressive knowledge aggregation. MultiCube-RAG [146] uses an ontology-guided cube structure to organize multi-hop retrieval without relying solely on noisy graph expansion. ReSearch [15] learns to reason with search via reinforcement learning under sparse outcome rewards. O1-Searcher [184] and similar systems explore deliberate search in a long-thought regime. Their advantage is adaptive depth; their risk is query drift, redundant calls, and weak attribution of which action improved observability.

5.2 Estimating Conditional Evidence Utility

A chain can be present in L_N but absent from E_K because the selection model optimizes marginal relevance rather than conditional utility. Rerankers, bridge selectors, compression methods, and plan-guided selectors therefore target $\theta_{\text{util}}(e \mid q, C)$.

5.2.1 Cross-encoder and LLM Rerankers. Classical rerankers [120] and stronger neural rerankers including MONOT5 [121], RANKT5 [233], RANKLLAMA [102], RANKGPT [153], RANKZEPHYR [128], and listwise LLM rerankers [160] score candidate documents by query relevance. In multi-hop settings, these scores are often misspecified: bridge documents can have low standalone relevance but high value after a partial chain has been observed. Some recent listwise LLM rerankers attempt joint scoring and are therefore closer to conditional utility estimation, but the gap from true conditional utility is rarely measured.

5.2.2 Bridge-aware and Core-conditioned Selection. Methods that condition selection on partial state begin to estimate true conditional utility. BALEEN's condensed retrieval [74] explicitly tracks accumulated evidence and reranks under that condition. HOPRETRIEVER [83] and PATHRETRIEVER [1] score candidate next-hop documents conditioned on entity or sentence state. BEAMDR [225] performs beam-search over chain hypotheses. MUGER2 [205] explicitly detects bridge entities. BRIDGE-R [200] adds a bridge-prediction objective to dense retrieval.

5.2.3 Compression and Note-based Methods. Compression can be interpreted as utility estimation under a token budget. LONGLLMLINGUA [65], and LONGLLMLINGUA-2 [125] compress prompts by predicting token importance. RECOMP [199] compresses retrieved passages into extractive or abstractive notes. CHAIN-OF-NOTE [216] writes intermediate notes that survive into the reader prompt. xRAG [16] compresses retrieved context into single tokens for efficient augmentation. The key statistical question for compression is whether chain-relevant complementary content survives, not whether the compressor preserves average relevance.

5.2.4 Plan-driven and Sub-question Selection. Sub-question and plan-based methods select evidence aligned with explicit reasoning steps. DECOMPRC [111], SELF-ASK [129], LEAST-TO-MOST [227], BREAK [189], PLAN-AND-SOLVE [175], HIPRAG [170], PROGRAMFC [123], and PAR-RAG's plan stage all transform a complex question into sub-tasks whose matched evidence is more nearly single-hop. The estimand interpretation is that decomposition reduces the variance of utility estimation: a sub-question constrains conditional context C explicitly, so each candidate document can be scored under a clearer conditional.

5.2.5 Training Signals for Utility. Conditional utility can be supervised in several ways: gold supporting facts [164; 204]; oracle ablations producing leave-one-out labels; reader-derived utility scores; verifier scores [155; 202]; and counterfactual deletion [96]. Each signal has bias. Gold supports may not be unique. Reader-derived utility can inherit reader-specific preferences. Verifier scores may reward plausibility rather than causal support. A robust selector should ideally combine multiple signals while reporting which utility surrogate is used.

From this view, the central statistical issue is selection bias under budget. A top-K context is not a neutral sample of the retrieved pool; it is a biased estimator of the latent support chain. Future selection models should therefore report not only relevance metrics but also conditional chain preservation: whether the missing supports of $C^\star$ survive the transition from L_N to E_K.

5.3 Estimating Exposure Probability

Exposure methods assume the relevant evidence is selected but not necessarily usable. Long contexts and multi-document prompts create position-dependent access problems [98; 214].

5.3.1 Reader-aware Ordering and Packing. Reader-aware ordering places bridge and answer-bearing evidence earlier or closer together. LongChat-aware packing strategies, "lost-in-the-middle" mitigations [98], LongContextBench-driven layout heuristics [4], and explicit instruction templates that announce chain structure are examples of low-cost ordering techniques. Some methods learn the ordering: PRCA [206] trains a context-aware adapter that re-orders evidence based on reader gradients. Position-Robust Context [53] fine-tunes LLMs to be position-invariant.

5.3.2 Fusion Architecture as Exposure. Architectures that decouple encoding from fusion reduce encoder-side interference. FiD [57] separately encodes passages and fuses them in the decoder, making each passage's contribution position-independent in the encoder. FiD-Light [56] and KG-FiD [215] extend the same idea with efficiency or structural priors. Atlas [58] co-trains retriever and FiD reader. Encoder-decoder separation is itself an exposure intervention because the reader no longer faces a single concatenated long prompt.

5.3.3 Compression-aware Exposure. Compression reduces exposure competition by shortening distractors. LLMLingua [65], RECOMP [199], and Selective-Context [88] all shrink the input and indirectly improve the reader's effective access depth. In fixed membership experiments, compression can isolate the exposure channel because it changes neither $\mathcal{S}$ nor the underlying retrieval.

5.3.4 Fixed-Membership Ordering as a Causal Probe. A method can preserve evidence membership while still improving answer quality by changing presentation. Fixed-membership ordering ablations where only the order of E_K varies, holding $|E_K|$ and contents constant, isolate θ_{exp}. Section 11 recommends this experiment as a standard supplement.

5.4 Estimating Fusion Reliability

Fusion reliability asks whether the reader can compose the chain once it is selected and exposed.

5.4.1 Retrieve-then-Read Readers. Retrieve-then-read systems such as DrQA [8], ORQA [80], RAG [82], REALM [48], Atlas [58], and FiD [57] are direct fusion estimators. They differ in whether retrieval is fixed, jointly trained, or distilled, and in whether passages are concatenated or independently encoded. Their shared role is to estimate $P(\hat{a} \mid q, E)$.

5.4.2 Graph-aware Fusion. Graph-aware fusion methods such as QA-GNN [212], GreaseLM [217], UniKGQA [67], ReaRev [108], SR [218], HamQA [23], and NSM [52] provide structured inductive bias for cross-evidence composition. The graph constrains how evidence units can be combined and therefore reduces the hypothesis space the reader must search over.

5.4.3 Decomposition and Programmatic Reasoning. Decomposition-based methods factorize a difficult multi-hop query into easier subproblems. DecompRC [111], Self-Ask [129], Least-to-Most [227], BREAK [189], Plan-and-Solve [175], Decomposed Prompting [75], Successive Prompting [24], and ReasonPath [87] all reduce fusion complexity by making intermediate operations explicit. Programmatic methods such as PoT [12], PAL [36], Faithful CoT [100], and Code-as-Reasoning [179] make intermediate operations executable or symbolically checkable. The main fusion-side failure is error propagation: a wrong early sub-answer corrupts later retrieval and reasoning, an effect amplified by the $\prod(1 - \epsilon_t)$ form of equation in §4.13.

5.4.4 Self-Consistency and Diverse Reasoning. Sampling-based methods improve fusion through diversity. Self-Consistency [176] marginalizes over multiple reasoning traces. Tree-of-Thought [210] and Graph-of-Thought [5] expand the reasoning topology beyond linear chains. Algorithm-of-Thought [142] introduces algorithmic templates. Auto-CoT [219] automatically constructs in-context exemplars. These methods do not change retrieval but increase the probability that at least one trajectory composes the chain correctly.

5.4.5 Latent and Compressed Reasoning Inside Readers. A separate fusion thread compresses reasoning into hidden states. Quiet-STaR [41], Coconut [51], pause-token methods [40], and "think before speak" variants generate internal computation that is not emitted as tokens. In the latent-chain view, these methods do not eliminate fusion error; they relocate it to model internals where it is harder to attribute. Empirically they can improve multi-hop accuracy on chain-explicit tasks [180] while making external diagnostics less informative.

5.5 Estimating Causal Faithfulness

Faithfulness methods estimate whether the answer is causally grounded in evidence rather than merely correct.

5.5.1 Self-verification. Self-verification methods use model-internal consistency or critique. Self-Refine [104], Reflexion [147], CoVe [21], SelfCheckGPT [107], and Self-Verification [188] score intermediate or final outputs by re-asking the model. They are cheap but bounded by the model's own knowledge and biases.

5.5.2 Externally-grounded Verification. Externally-grounded methods retrieve or check additional evidence before revising outputs. CRITIC [39], RARR [37], Verify-and-Edit [226], CRAG [203], Self-Checker [91], and Attributable-Trace [84] ground critique in retrieved content. Self-RAG [2] combines retrieval and critique tokens in a single decoding loop.

5.5.3 Process Reward Models. Process-supervised reward models shift faithfulness from post-hoc evaluation to a training signal. Lightman PRM [94], Math-Shepherd [174], ReARTeR [155], RAG-Gym [197], and VERITAS [202] reward intermediate reasoning steps or evidence-grounded actions rather than only final correctness. ReST-MCTS [223] and similar systems combine search and process supervision.

5.5.4 Counterfactual and Deletion Tests. Counterfactual benchmarks such as CofCA [190] test whether changing a bridge variable changes the answer. Deletion tests [96; 201] remove a candidate supporting passage and measure whether the answer changes. Both probe θ_{faith} directly because they intervene on evidence rather than asking the model to reflect.

5.5.5 Citation and Attribution Models. Citation-aware generators WebGPT [113], GopherCite [110], and ALCE [33] produce per-claim citations that can be validated for entailment. Citation precision is necessary but not sufficient for faithfulness, as a citation can be textually relevant without being causally used.

Under the estimand view, all of these methods target Δ_{faith}. They differ in whether they intervene pre-hoc, in-decoding, or post-hoc, and in whether the supervision signal is internal, external, or interventional. We return to causal faithfulness in detail in Section 9.

5.6 Joint Estimation in Agentic and Hybrid Systems

Hybrid and agentic systems estimate several quantities simultaneously. Self-RAG [2] decides when to retrieve, critiques evidence, and generates. Search-R1 [71] and DeepRAG [43] learn retrieval and reasoning policies with RL. LM–KG hybrids such as ToG 2.0 [103], RoG [99], KAG [93], HippoRAG [47], LightRAG [46], SubgraphRAG [92], and GraphRAG [25] combine structured chain priors with flexible language reasoning. Multi-agent systems such as MA-RAG [115], ReAgent [185], and Chain-of-Agents [220] distribute decomposition, retrieval, reasoning, and verification across roles.

Table 5. Recent 2025–2026 multi-hop RAG methods interpreted by primary estimand.

Method	Mechanism	Primary estimand	Statistical interpretation
Search-R1 [71]	RL-trained search-query generation	Observability / control	Learns a policy over retrieval actions that increases probability of useful external evidence.
DeepRAG [43]	MDP over retrieve-or-reason decisions	Cost-aware observability	Trades retrieval calls against expected information gain at each reasoning step.
R1-Searcher [149]	RL-incentivized search via reward shaping	Observability	Improves bridge discovery through intermediate retrieval reward.
ReSearch [15]	RL search under outcome-only reward	Observability / control	Demonstrates that pure RL can yield effective search agents.
RAG-Gym [197]	Process supervision for search agents	Observability and utility	Provides intermediate rewards for search actions rather than only final-answer reward.
PAR-RAG [224]	Plan-act-review with verification	Fusion and faithfulness	Controls reasoning drift by reviewing intermediate results.
DualRAG [17]	Reasoning-augmented querying plus knowledge aggregation	Utility / fusion	Couples query generation with progressive evidence integration.
VERITAS [202]	Faithfulness-aware process rewards	Causal faithfulness	Optimizes reasoning traces for entailment and support, not just correctness.
ReARTeR [155]	Trustworthy PRM for retrieval-augmented reasoning	Faithfulness / fusion	Process-level reward model trained for retrieval-grounded steps.
MultiCube-RAG [146]	Ontology-guided cube retrieval	Structural observability	Restricts retrieval to semantically organized dimensions to reduce graph noise.

The advantage of joint estimation is end-to-end adaptability. The cost is attribution: when answer accuracy improves, it is hard to tell whether the gain came from observability, selection, exposure, fusion, or faithfulness. We therefore recommend that future agentic and hybrid systems report component-level diagnostics aligned with the five estimands rather than only final answer scores.

5.7 When Method Families Collide

Recent systems increasingly combine several families: a method may use a planner, a dense retriever, a graph index, a long-context reader, and a verifier. Architectural labels become less informative in such hybrids. The estimand view avoids this problem by decomposing a hybrid into the quantities it targets. For example, a graph-augmented agent may use graph neighbors to improve observability, an LLM planner to improve conditional utility, a context packer to improve exposure, and a verifier to improve faithfulness. The method should then be evaluated at each link rather than described only as "GraphRAG" or "agentic RAG."

By Theorem 1, a strong retriever may not improve answer accuracy when the reader cannot fuse evidence. A strong reader may not help when the chain is absent from the pool. A faithfulness verifier may reject bad answers but not recover missing evidence. Therefore the most credible empirical papers are those that show where in the product the gain occurs.

Table 6. Architectural families mapped to latent-chain estimands. A checkmark indicates a primary contribution; (◦) indicates a secondary contribution.

Family	Observability	Utility	Exposure	Fusion	Faithfulness
Retrieval-based	✓	(◦)			
Graph-based	✓	(◦)		✓	✓
Decomposition-based	(◦)	(◦)		✓	(◦)
Fusion-based readers			(◦)	✓	
LLM-based reasoning	(◦)	(◦)	(◦)	✓	(◦)
Verification-based				(◦)	✓
Long-context / hierarchical	(◦)	(◦)	✓	(◦)	
Hybrid / agentic	✓	✓	(◦)	✓	✓

6 Architectural Families: A Complementary Review

The estimand-based taxonomy does not replace architectural taxonomies; it explains why they are incomplete. This section provides a parallel architectural review for readers familiar with prior surveys. Each family is reviewed with its historical development, representative methods, primary failure modes, and the estimands it most directly affects. Table 6 provides a compact crosswalk.

6.1 Graph-based Methods

Graph methods make paths first-class. Their strongest contribution is structured fusion and faithfulness: a path can be audited hop by hop.

Lineage 1: Question-specific entity or document graphs. GRAFT-Net [151] fuses heterogeneous text and KG nodes for early-fusion question answering. PullNet [152] iteratively pulls neighbors to construct a question-specific subgraph. DFGN [133] performs dynamically fused graph reasoning over entity graphs constructed from supporting paragraphs. HGN [27] stacks hierarchical entity-, sentence-, and paragraph-level nodes. SAE [168] adds adaptive selection and explanation. HDE [167] encodes heterogeneous document edges.

Lineage 2: LLM-coupled graph reasoners. QA-GNN [212] couples a LM with a GNN over a question-induced subgraph. GreaseLM [217] fuses LM and GNN representations layer by layer. UniKGQA [67] unifies retrieval and reasoning on KGs. ReaRev [108] performs representation-iterative reasoning over relational paths. KG-FiD [215] infuses KG signals into a FiD reader. NSM [52] learns a neural state machine for multi-hop KGQA. EmbedKGQA [138] uses knowledge graph embeddings for multi-hop reasoning over incomplete graphs. TransferNet [145] reasons over relation graphs by transferring activation between entity sets.

Lineage 3: LLM-built and LLM-traversed graphs. ToG [154] and ToG 2.0 [103] traverse KG paths under LLM control. RoG [99] grounds LLM reasoning on retrieved relational paths. GraphRAG [25] builds an LLM-extracted graph index and uses community-level summaries for query-focused generation. LightRAG [46] offers a cost-efficient alternative. HippoRAG [47] provides hippocampus-inspired long-term memory through personalized PageRank over an LLM-built graph. SubgraphRAG [92] retrieves task-relevant subgraphs. KGP [181] performs knowledge graph prompting on documents.

Estimand role. Graphs provide a prior over possible chains. They improve observability when graph expansion reaches bridge nodes; they improve fusion when paths restrict how evidence units can be combined; and they improve faithfulness when each hop can be audited. Their limitation is graph construction quality. Missing nodes, noisy entity linking, and incomplete edges reduce observability before the reasoner is invoked. LLM-built graphs reduce manual engineering but move the bottleneck to extraction reliability.

6.2 Retrieval-based Methods

Retrieval methods are the direct response to evidence missingness. They span single-shot, iterative, adaptive, and hierarchical variants.

Single-shot retrieval. BM25 [135] remains a strong baseline. DPR [72] introduced dense passage retrieval. ColBERT [73] and ColBERTv2 [136] use late interaction. SPLADE [32] fuses sparse and learned expansion. ANCE [194] introduced approximate nearest neighbor negative contrastive estimation. Contriever [59] pretrains retrievers contrastively without labels. GTR [116], E5 [173], BGE [193], and RetroMAE [192] provide stronger general embeddings. HyDE [35] synthesizes hypothetical documents to bridge the lexical gap.

Iterative and reasoning-interleaved retrieval. GoldEn [112], MDR [195], Baleen [74], MUPPET [28], IRCoT [165], FLARE [64], Iter-RetGen [144], ITRG [30], and DRAGIN [150] condition retrieval on accumulated evidence or reasoning.

Adaptive and selective retrieval. Self-RAG [2], Adaptive-RAG [60], SKR [177], When-to-Retrieve [106], and In-Context RAG [134] decide whether and when retrieval is needed.

Long-context and hierarchical retrieval. LongRAG [68], RAPTOR [137], HiQA [13], Memory-RAG [130], and LongContextBench-aligned systems [4; 214] reduce the number of explicit hops by changing retrieval granularity.

Robust and corrective retrieval. CRAG [203] adds a corrective retrieval loop with quality scoring. Self-Refine RAG [90] iteratively refines retrieved evidence. Active RAG [89] introduces an active learning component.

Estimand role. The core tradeoff is coverage versus noise. Iterative, reasoning-interleaved, and adaptive retrieval all increase chain observability, but they introduce query drift and distractor mass. Their success should therefore be measured by both pool-chain coverage and noise ratio, not recall alone.

6.3 Decomposition-based Methods

Decomposition methods expose the dependency structure of q and reduce fusion complexity.

Sub-question methods. DecompRC [111], Self-Ask [129], BREAK [189], Least-to-Most [227], Decomposed Prompting [75], Successive Prompting [24], and Plan-and-Solve [175] rewrite a hard question into a sequence of simpler queries or operations.

Programmatic decomposition. PoT [12] and PAL [36] generate executable programs. Faithful CoT [100] separates planning from execution. ProgramFC [123] generates verification programs. Self-Discover [229] chooses reasoning modules adaptively.

Plan-driven decomposition. PAR-RAG [224] adopts a plan-act-review loop. HiPRAG [170] performs hierarchical planning for multi-hop QA. Plan*RAG [18] uses planner-guided retrieval.

Estimand role. Decomposition changes both observability and fusion reliability. It improves observability when sub-questions become more targeted retrieval queries, and it improves fusion when intermediate operations reduce the complexity of composition. Its bottleneck is cascading error: once an intermediate sub-question is wrong, later evidence estimates become biased. Granularity is also difficult to tune: coarse decompositions do not reduce multi-hop difficulty, whereas overly fine decompositions increase latency and error accumulation.

6.4 Fusion-based Readers

Fusion methods address the conditional probability that a reader answers correctly given selected evidence.

Retrieve-then-read backbones. DrQA [8], ORQA [80], RAG [82], REALM [48], Atlas [58], FiD [57], FiD-Light [56], KG-FiD [215], and EMDR2 [148] establish the basic pattern.

Compression-aware fusion. RECOMP [199], LLMLingua [65], LongLLMLingua [66], Chain-of-Note [216], xRAG [16], and Selective-Context [88] reduce the amount of irrelevant content presented to the reader.

Long-context and memory-augmented readers. LongChat, LongLLaMA [169], Mamba-based RAG [42], and KV-cache-aware retrieval pipelines [14] represent the emerging direction of fusing retrieval and long-context architectures.

Estimand role. Fusion methods are downstream estimators. They can reduce noise and cross-passage interference, but they cannot repair missing evidence. They also cannot guarantee causal faithfulness: a reader can combine evidence correctly in one case and still rationalize from parametric memory in another. Fusion should therefore be evaluated under controlled evidence membership, ideally with oracle or near-oracle contexts.

6.5 LLM-based Reasoning Methods

LLM-based reasoning blurs architectural boundaries.

Linear chain-of-thought. CoT [186], Zero-shot CoT [76], Auto-CoT [219], and Self-Consistency [176] expand the reasoning capacity of LLMs through structured prompts.

Non-linear topologies. Tree-of-Thought [210], Graph-of-Thought [5], Algorithm-of-Thought [142], Skeleton-of-Thought [117], and Cumulative Reasoning [221] expand the expression space.

Programmatic reasoning. PoT [12], PAL [36], Faithful CoT [100], and Code-as-Reasoning [179] use executable code to anchor reasoning.

Latent reasoning. Quiet-STaR [41], Coconut [51], and Pause-Tokens [40] compress reasoning into hidden states.

Tool-use and agentic reasoning. ReAct [209], Toolformer [139], ToolLLM [132], and Gorilla [127] interleave tool calls with reasoning. Voyager [178] and Generative Agents [126] extend reasoning to embodied or social settings.

Search-based reasoning. ToT [210] uses BFS/DFS over thought trees. LATS [228] adds MCTS. Reasoning-via-Planning [50] introduces deliberate search.

Estimand role. This flexibility is powerful, but it makes evaluation entangled. A single policy may retrieve, decompose, reason, verify, and stop. If its final answer improves, the mechanism of improvement is not obvious. It may have learned better retrieval, better decomposition, better use of retrieved evidence, or better guessing. Estimand-level diagnostics are necessary to determine what the policy has actually learned.

6.6 Verification-based Methods

Verification methods estimate whether the answer is supported.

Self-verification. Self-Refine [104], Reflexion [147], CoVe [21], SelfCheckGPT [107], and Self-Verification [188] rely on model-internal critique.

External verification. CRITIC [39], RARR [37], Verify-and-Edit [226], and CRAG [203] ground critique in retrieval.

Process reward models. Lightman PRM [94], Math-Shepherd [174], ReARTeR [155], RAG-Gym [197], and VERITAS [202] provide step-level training signal.

Counterfactual evaluation. CofCA [190] and related benchmarks [96; 201] test causal use.

Estimand role. Verification primarily targets faithfulness. Its limitation is that verification cannot repair a missing chain; it can only reject a faulty one or trigger another retrieval round.

6.7 Long-context and Multimodal Multi-hop Reasoning

Two recent directions stress different substrates of $\mathcal{K}$.

Long-context reasoning. Lost-in-the-middle effects [98] motivate positional-bias-aware reading. LongBench [4], ∞Bench [222], and NovelHopQA [122] stress long-context multi-hop behavior. LongLLaMA [169] and Mamba [42] extend context length through architectural changes. LongRAG [68] and RAPTOR [137] adapt retrieval granularity.

Multimodal multi-hop. MultiModalQA [159], WebQA [7], ManyModalQA [49], MMRA [81], VisDoM [183], DocHop-QA [156], and WikiMixQA [54] require alignment across text, tables, and images. Img2LLM [45] and LLaVA-RAG [95] build multimodal RAG pipelines.

Tabular and semi-structured. HybridQA [9], OTT-QA [11], TabFact [10], FeTaQA [114], and TAT-QA [231] require composition across cells and linked passages.

6.8 Hybrid and Agentic RAG

Hybrid and agentic systems combine several families. Self-RAG [2] couples retrieval, critique, and generation. Search-R1 [71], R1-Searcher [149], ReSearch [15], and DeepRAG [43] train policies via RL. MA-RAG [115], ReAgent [185], and Chain-of-Agents [220] use multi-agent orchestration. KAG [93], HippoRAG [47], GraphRAG [25], LightRAG [46], and SubgraphRAG [92] combine KGs and LLMs.

The attribution problem in joint policies is acute. By Theorem 1, any improvement in final answer accuracy must come from improving at least one of the five factors. Without per-factor diagnostics, a hybrid system's improvement is not reproducible across architectural choices.

6.9 Cross-cutting Mechanism Analyses

A small but growing literature studies the *mechanism* by which LLMs perform multi-hop reasoning. Dissecting multi-hop [38] traces how intermediate facts are recalled and combined. Do LLMs Latently Perform Multi-Hop Reasoning? [208] finds that some models recall bridge entities in middle layers but use them inconsistently. Hopping [6] characterizes step-by-step recall. Grokked Transformers [180] show that generalization on multi-hop tasks emerges with extended training and specific circuit formation. These analyses are not yet integrated with retrieval-augmented systems, but they speak directly to the question of whether the chain object exists inside the model. In the latent-chain view, mechanistic analyses provide a third source of information about the chain (in addition to retrieved evidence and external annotations) and may eventually support new faithfulness diagnostics.

7 Benchmarks as Diagnostic Instruments

Benchmarks differ less by domain than by which latent event they make observable. We organize benchmarks by diagnostic function rather than by dataset family. A useful benchmark, under the latent-chain view, exposes at least one of the five estimands while controlling for the others.

7.1 Coverage and Chain-Observability Diagnostics

HotpotQA [204], 2WikiMultiHopQA [55], IIRC [31], ComplexWebQuestions [158], QAngaroo [187], and open-domain variants of multi-hop QA primarily test whether systems can retrieve enough evidence to make the latent chain observable. Supporting-fact labels provide partial observability of $\mathcal{P}$ and $\mathcal{S}$, although they do not always identify the chain actually used by the model. The classic weakness of HotpotQA is that many examples can be answered correctly from a single paragraph due to lexical artifacts; the "distractor setting" provides better selection diagnostics than the "full-wiki setting," but neither fully isolates membership from ordering effects.

7.2 Shortcut-Resistance Diagnostics

MuSiQue [164] was designed by composing single-hop questions and filtering shortcuts, making it a stronger test of true chain observability than datasets where one paragraph may suffice. MoreHopQA [141] stresses longer chains, where observability compounds multiplicatively. NovelHopQA [122] uses long-form narrative sources where chain reasoning interacts with long-context layout. HotpotQA-NoShortcut [163] provides a filtered subset that requires real multi-hop. These benchmarks are valuable because they reduce the chance that high answer accuracy comes from shallow cues rather than successful chain inference.

7.3 Path and Process Diagnostics

2WikiMultiHopQA supplies explicit reasoning paths. EntailmentBank [19] provides entailment trees over scientific facts. eQASC [61] provides explanation chains over commonsense reasoning. ProofWriter [157] provides synthetic proofs. HoVer [63], EX-FEVER [101], and FoolMeTwice [26] annotate multi-step evidence for verification. These benchmarks help diagnose $\mathcal{U}$, because they make some intermediate structure visible rather than evaluating only final answers.

7.4 Heterogeneous-Evidence Diagnostics

CWQ [158], HybridQA [9], OTT-QA [11], MultiModalQA [159], WebQA [7], ManyModalQA [49], TabFact [10], TAT-QA [231], FeTaQA [114], VisDoM [183], DocHop-QA [156], and WikiMixQA [54] stress fusion across structured and unstructured sources. Their diagnostic value lies less in raw answer accuracy than in whether a method can maintain the chain across different evidence types.

7.5 Counterfactual and Faithfulness Diagnostics

CofCA [190], citation-faithfulness benchmarks ALCE [33], HaluEval [86], ExpertQA [105], and fact-verification datasets such as FEVER [162], HoVer [63], EX-FEVER [101], SciFact [171], and AVeriTeC [140] probe whether the answer changes when bridge variables or supporting facts are perturbed. These benchmarks are closest to measuring θ_{faith}, because they ask whether evidence is causally load-bearing rather than merely cited.

Table 7. Representative benchmark families and the latent event they most directly expose.

Benchmark family	Examples	Primary estimand	Main caution
Supporting-fact QA	HotpotQA, 2WikiMultiHopQA, IIRC, ComplexWebQuestions	Observability and selection	Shortcut artifacts may inflate answer accuracy.
Composed / shortcut-resistant QA	MuSiQue, MoreHopQA	Bridge observability	Lower scores may reflect pool absence rather than reader weakness.
Path-annotated QA	2WikiMultiHopQA, EntailmentBank, eQASC, ProofWriter	Path validity / fusion	Gold path may not be unique.
Heterogeneous evidence	HybridQA, MultiModalQA, OTT-QA, WebQA, TAT-QA, FeTaQA, VisDoM	Fusion reliability	Alignment errors can dominate retrieval errors.
Long-context multi-hop	LongBench, ∞Bench, HELMET, NovelHopQA	Exposure	Membership and order are not separately controlled by default.
Fact verification	FEVER, HoVer, EX-FEVER, SciFact, AVeriTeC	Faithfulness and support validity	Veracity labels do not always identify generation causality.
RAG-specific diagnostics	MultiHop-RAG, FRAMES, FanOutQA, RAGTruth, CRAG-bench	End-to-end RAG behavior	Requires separating retrieval, selection, citation, and answer quality.
Counterfactual QA	CofCA and related perturbation sets	Causal faithfulness	Expensive annotation and possible unnatural perturbations.
Agentic / tool use	ToolBench, AgentBench, WebArena	Joint observability and control	Difficulty depends on tool affordances.

7.6 Long-context Multi-hop Diagnostics

LongBench [4], ∞Bench [222], HELMET [213], NovelHopQA [122], NarrativeQA [77], Loong [182], and ZeroSCROLLS [143] stress chain construction under long context, where exposure failures dominate over selection failures.

7.7 RAG-specific and Agentic Diagnostics

MultiHop-RAG [161], FRAMES [78], CRAG [207], FanOutQA [232], and RAGTruth [118] evaluate end-to-end RAG behavior under multi-step queries. ToolBench [131] and AgentBench [97] test agentic search behavior. WebArena [230] introduces realistic agent-environment interaction.

7.8 Metric Alignment

A metric should be matched to the estimand it claims to measure. Answer EM and F1 are deployment-level metrics, not mechanism-level metrics. Supporting-fact F1 approximates evidence membership but does not establish causal use. Retrieval recall at Top-N estimates pool observability but says little about the final reader context. Citation precision estimates attribution correctness but can be high even when the answer was produced from parametric memory. Counterfactual sensitivity is closer to causal faithfulness but depends on the quality of the intervention.

Table 8. Metric–estimand alignment. A strong paper should avoid using a metric from one row as evidence for a claim in another row without an explicit caveat.

Claim	Suitable metric	Conditional subset	Misleading shortcut
Pool contains the chain	support/path recall at Top-N	all examples or by hop count	Answer F1 alone.
Budget preserves the chain	selected support/path recall at K	chain-present in pool	Reranker relevance score.
Ordering helps the reader	early support exposure; fixed-set order ablation	selected-chain-present	Changing membership and order together.
Reader can compose evidence	oracle-evidence EM/F1; step accuracy	oracle or selected-chain-present	Top-K retrieval recall.
Answer is faithful	deletion sensitivity; counterfactual support tests; citation entailment	answer-correct subset	Citation overlap alone.
System is efficient	latency, token cost, retrieval calls, verifier calls	same answer-quality band	Selector-only latency without reader cost.

7.9 Dataset Families Reinterpreted

Traditional benchmark descriptions emphasize source domain and release year. The estimand view emphasizes which latent event a benchmark makes measurable. HotpotQA is valuable because sentence-level support labels enable pool and context diagnostics, but its shortcut artifacts mean that answer accuracy can overstate chain use. 2WikiMultiHopQA adds explicit path structure, making it useful for path-level observability and fusion analysis. MuSiQue is valuable because it reduces single-hop shortcuts and stresses genuine bridge observability. IIRC makes missing context explicit, placing retrieval actions at the center. FEVER, HoVer, and EX-FEVER foreground support validity rather than answer-string matching. HybridQA and MultiModalQA stress fusion because a correct chain may cross tables, linked passages, and other semi-structured sources.

This reinterpretation changes how benchmark coverage should be reported. A paper that evaluates only on shortcut-prone datasets primarily shows that the system can exploit available signals; it does not establish robust latent-chain inference. A paper that evaluates only on verification datasets may test faithfulness without testing open-domain chain observability. A strong empirical section should therefore include at least one benchmark for observability, one for shortcut resistance, and one for faithfulness or counterfactual sensitivity.

8 Applications

Applications differ by which estimand dominates the deployment risk.

8.1 Open-Domain Multi-Hop QA

Open-domain QA is dominated by observability and budget risk. Iterative or agentic retrieval improves $P(\mathcal{P})$ but increases latency and noise. A deployable system must therefore learn when additional retrieval is worth its marginal information gain. Practical systems such as Self-RAG [2], Adaptive-RAG [60], Search-R1 [71], and DeepRAG [43] represent different tradeoff points on the depth–cost frontier.

8.2 Knowledge-Graph Question Answering

KGQA offers structured paths and strong faithfulness, but coverage is bounded by graph completeness. Hybrid text–graph methods such as KAG [93], GraphRAG [25], HippoRAG [47], ToG 2.0 [103], and RoG [99] improve observability while preserving a structured prior over chains. Industrial deployments often favor KG-anchored pipelines because path-level faithfulness is auditable.

8.3 Fact Verification and Claim Checking

Fact verification foregrounds causal faithfulness. The output is often a label rather than a generated answer, but the latent-chain problem is the same: identify evidence, compose it, and verify that the claim is supported. Benchmarks such as FEVER [162], HoVer [63], EX-FEVER [101], SciFact [171], and AVeriTeC [140] test multi-step support; ProgramFC [123] casts verification as program execution; ClaimReview-anchored systems [3] provide real-world deployment context.

8.4 Dialogue and Interactive Systems

Dialogue introduces conversation history as an additional state. The system must decide whether the missing link should be retrieved, inferred from history, or requested from the user. This makes observability and action controllability central. Conversational RAG systems such as Wizard of Wikipedia [22], DialDoc [29], and Chat-RAG [198] extend the framework to multi-turn settings.

8.5 Legal, Medical, and Scientific Reasoning

High-stakes domains amplify faithfulness and heterogeneous-fusion risk. Evidence may include statutes, cases, guidelines, records, papers, tables, and figures. These settings favor explicit chains, component attribution, and counterfactual verification. Legal QA benchmarks such as LegalBench [44], medical QA such as MedQA [70], PubMedQA [69], BioASQ [166], MIRAGE [196], and scientific QA such as QASPER [20] and SciQAG [172] all reward systems that maintain visible support chains.

8.6 Enterprise and Customer-Support RAG

Enterprise RAG often involves policies, product documentation, tickets, and historical conversations. The dominant risk is not only answer accuracy but traceability: the system must show which documents support a recommendation and whether the support chain crosses multiple internal sources. Conditional utility is important because a short policy clause may be decisive only after a user-specific constraint has been identified. Systems based on GraphRAG [25] and LightRAG [46] are widely deployed because graph indexes simplify attribution.

8.7 Scientific Literature Assistance

Scientific assistance requires chaining claims across papers, methods, datasets, tables, and sometimes code. Observability is difficult because terminology varies across subfields. Fusion is difficult because the system must compare assumptions, experimental settings, and measurement protocols. Faithfulness is crucial because a correct-sounding synthesis without exact support is harmful. These settings motivate citation-level and claim-level counterfactual checks, and have been explored by PaperQA [79], ScisummNet-style summarization with multi-hop support [211], and HiQA [13].

Analysis and Outlook

Analysis and Outlook

9 Faithfulness as Causal Evidence Use

We argued in Section 4.5 that faithfulness is a causal estimand. A high-quality multi-hop RAG system should produce answers that are causally supported by the retrieved chain rather than answers that merely co-occur with plausible-looking citations. This section develops the causal view in detail, connects it to existing faithfulness diagnostics, and explains why several common metrics are necessary but not sufficient. We also use the running example (Section 1.1) to illustrate each failure mode.

9.1 A Causal Graph for Multi-Hop RAG

Let W denote the world state, $\mathcal{K}$ the knowledge store derived from W, $C^\star$ the latent support chain, L_N the retrieved pool, E_K the selected context, $\pi(E_K)$ the ordering, and $\hat{a}$ the answer. A simple causal graph for a faithful system is

$$W \to \mathcal{K} \to C^\star \to L_N \to E_K \to \pi(E_K) \to \hat{a}, \tag{42}$$

with the parametric memory of the reader Θ acting as an exogenous influence on $\hat{a}$:

$$\Theta \to \hat{a}. \tag{43}$$

A causally faithful answer is one whose distribution responds to interventions on $C^\star$ through $\pi(E_K)$, i.e.

$$P(\hat{a} \mid \mathrm{do}(\pi(E_K) = \pi')) \neq P(\hat{a} \mid \pi(E_K)) \quad \text{whenever} \quad P(a^\star \mid \pi(E_K)) \neq P(a^\star \mid \pi'). \tag{44}$$

An unfaithful answer can be partitioned into two cases: *parametric leakage*, in which $\hat{a}$ is generated through Θ alone and is invariant under context interventions; and *shortcut use*, in which $\hat{a}$ depends on a non-evidence feature of $\pi(E_K)$ such as template wording, length, or surface lexical overlap.

9.2 Deletion and Counterfactual Sensitivity

Two related interventions probe causal use.

Deletion sensitivity. For each selected evidence unit $e \in E_K$, define

$$S_{\mathrm{del}}(e) = P(\hat{a} = a^\star \mid \pi(E_K)) - P(\hat{a} = a^\star \mid \pi(E_K \setminus \{e\})). \tag{45}$$

A faithful answer should exhibit a positive deletion sensitivity on support units of the latent chain. Sufficient evidence metrics [96] and the Attributable-Trace framework [84] implement variants of this idea.

Counterfactual bridge sensitivity. Let $e_b^\star$ be a bridge unit whose factual content can be altered to obtain a counterfactual version e_b' encoding a different but consistent alternative (e.g., a different country in the running example). Counterfactual benchmarks [190] test whether the answer shifts accordingly. A reader that answers correctly regardless of e_b' is exploiting parametric memory; a reader whose answer shifts to remain consistent with the counterfactual bridge is using the chain causally.

9.3 Correctness, Citation, and Faithfulness Are Distinct

A common methodological mistake is to treat citation correctness and answer correctness as faithfulness. They are necessary signals but not sufficient.

Remark 2 (Three independent failure modes). For each example, the joint outcome ($\hat{a} \overset{?}{=} a^\star$, citations valid?, $\hat{a} \overset{?}{\leftarrow} C^\star$) takes one of eight truth-value combinations. Six of the eight are pathological. Of particular interest are: correct answer with valid citation but no causal use (parametric leakage); correct answer without valid citation (citation suppression); valid citation with wrong answer (citation without composition). A single number such as answer F1 or citation precision cannot distinguish these cases.

A practical recommendation follows: faithfulness audits should report all three signals jointly, or at minimum stratify answer correctness by citation validity and by deletion sensitivity. Papers that report only one of the three numbers should be read with this caveat in mind.

9.4 Shortcut Exploitation and Parametric Leakage

Shortcut and leakage failures are not equally easy to detect. Shortcut exploitation usually leaves a measurable trace: removing a shortcut feature (such as the surface form of an answer entity in the question) changes accuracy. MuSiQue [164] and shortcut-filtered subsets of HotpotQA [163] were constructed precisely to reduce this bias.

Parametric leakage is harder to detect because the leaked answer is plausibly produced by the chain. Closed-book and oracle-evidence ablations are useful but incomplete: closed-book accuracy estimates parametric availability, but a model can still leak even when oracle evidence is provided. Counterfactual benchmarks are the closest approximation, because they make leakage observable through the divergence between counterfactual chain and observed answer.

9.5 From Verification to Training Signal

Faithfulness has historically been treated as a post-hoc evaluation. Recent work moves it upstream into training. Math-Shepherd [174] and the process reward model of Lightman et al. [94] reward intermediate reasoning steps rather than only final answers. ReARTeR [155] trains a trustworthy process reward model tailored to retrieval-augmented reasoning. VERITAS [202] uses faithfulness-aware reward signals. RAG-Gym [197] adds process supervision to search agents. PAR-RAG [224] inserts a review stage that catches drift before it propagates.

From the latent-chain view, all of these methods convert faithfulness from an evaluation property into an objective. They differ in whether the reward is supplied by humans, by a verifier, or by self-critique; in whether the reward is per-step or per-trajectory; and in whether the training data includes interventional examples. Combining process supervision with counterfactual evaluation appears to be one of the most promising directions for measurable faithfulness gains.

9.6 Failure Modes Revisited

The five failure events in Section 3.3—evidence absence, budgeted selection failure, exposure failure, fusion failure, faithfulness failure—can now be paired with specific diagnostics.

10 Statistical Tradeoffs and Open Problems

The latent-chain view reorganizes open problems as tradeoffs among estimands. Each tradeoff is a constraint: improving one estimand often worsens another. A successful method does not eliminate the tradeoff; it moves the system to a better point on the frontier.

10.1 Eight Open Problems

OP1. Joint optimization of observability and noise. Iterative and agentic retrieval increase θ_{obs} but inflate ρ_N. No widely accepted method jointly minimizes expected risk under a budget. Open directions include retrieval policies trained against joint chain-coverage and noise-ratio objectives, and verifier-in-the-loop pruning that operates on the pool rather than on the reader's output.

Table 9. Failure events paired with diagnostics and recommended interventions.

Failure event	Diagnostic	Recommended intervention
Evidence absence	Pool support recall conditional on hop count	Iterative or graph-expansion retrieval; agentic search; coarser or hierarchical retrieval units.
Budgeted selection failure	Selected support recall under fixed K	Bridge-aware or core-conditioned selection; conditional-utility reranking.
Exposure failure	Order-only ablation on fixed E_K	Reader-aware ordering; compression; decoder-side fusion (FiD).
Fusion failure	Oracle-evidence reader accuracy	Decomposition; programmatic reasoning; structured graph fusion.
Faithfulness failure	Deletion and counterfactual sensitivity	Process reward; citation-aware decoding; counterfactual training.

OP2. Conditional utility estimation under chain dependence. Most rerankers estimate marginal relevance. Conditional utility estimation under partial chains is still rare and is hard to supervise. Open directions include core-conditioned scoring, bridge-classification objectives, and synthetic counterfactual chains for utility regression training.

OP3. Reader-aware exposure under long context. Long-context models have not eliminated positional bias [98; 214]. Open directions include fixed-membership ordering policies, decoder-side fusion (FiD variants), context compression, and explicit attention scaffolds.

OP4. Auditable latent reasoning. Latent reasoning methods [41; 51] trade chain visibility for inference efficiency. Faithfulness audits based on emitted tokens cease to apply. Open directions include mechanistic probes for internal bridge representations and hybrid latent–token reasoning where the chain is occasionally externalized.

OP5. Causal evaluation for retrieval-augmented systems. Counterfactual evaluation is expensive. Most benchmarks still use factual data only. Open directions include synthetic counterfactual chain generation, deletion-based attribution at the chain level rather than at the token level, and joint citation–causal-use metrics.

OP6. Knowledge update without catastrophic faithfulness loss. Retrieval is a natural mechanism for keeping knowledge current, but edits in $\mathcal{K}$ can conflict with parametric memory. Open directions include retrieval policies that detect knowledge-update events, and reader objectives that prefer retrieved evidence over parametric memory under conflict [106; 247].

OP7. Cost-aware multi-hop RAG.. Agentic and iterative methods increase latency and token cost. A deployable system must trade these against expected chain observability. Open directions include cost-sensitive policy learning, calibrated retrieval termination, and small-model verifiers paired with strong readers.

OP8. Safety and robustness under retrieval-side adversaries. Retrieval-side prompt injection and corpus poisoning can compromise otherwise faithful systems [248; 250; 251]. Open directions include trust-aware retrieval, multi-source agreement, and adversarial training for retrievers.

Table 10. Multi-hop RAG tradeoffs framed as estimand tensions.

Tension	Estimands	Typical methods	What an honest evaluation should show
Coverage vs. noise	$\theta_{\text{obs}} \uparrow$ vs. $\rho_N \uparrow$	Iterative / agentic retrieval	Both pool recall and pool noise on the same examples.
Budget vs. chain preservation	$K \downarrow$ vs. $\theta_{\text{util}} \uparrow$	Reranking, compression, bridge selection	Selected support recall under fixed K.
Exposure vs. budget	$\theta_{\text{exp}} \uparrow$ vs. $K \uparrow$	Reordering, packing, FiD	Fixed-membership ordering ablation.
Fusion power vs. latency	$\theta_{\text{fus}} \uparrow$ vs. cost$\uparrow$	Decomposition, programmatic reasoning, agents	Latency and tokens per question at fixed accuracy.
Plausibility vs. causal use	accuracy$\uparrow$ vs. $\Delta_{\text{faith}} \downarrow$	Verification, counterfactual training, PRM	Deletion and counterfactual sensitivity.
Controllability vs. adaptivity	step-level audit$\uparrow$ vs. policy freedom$\uparrow$	Multi-agent vs. end-to-end RL	Per-step diagnostics on the same model.

Table 11. Estimands, the closest existing surrogate metrics, and example estimator improvements suggested by the latent-chain view.

Estimand	Closest surrogate today	Improvement direction
θ_{obs}	Pool support recall at Top-N	Chain recall conditional on bridge gold; recall under controlled distractor density; agentic-action attribution.
θ_{util}	Marginal relevance scores	Core-conditioned utility; bridge prediction; oracle leave-one-out under partial chains.
θ_{exp}	Average position of support	Fixed-membership ordering EM gap; access-depth-conditioned accuracy; lost-in-middle test sets.
θ_{fus}	Oracle-evidence EM/F1	Step accuracy under explicit chains; program execution validity; intervention robustness.
θ_{faith}	Citation precision	Joint citation + deletion + counterfactual + parametric-leakage decomposition.

10.2 Tradeoffs Between Estimands

10.3 Estimator Suggestions for Future Work

11 Reporting Guidelines for Multi-Hop RAG

The latent-chain view directly implies a small set of reporting guidelines. We collect them here as a checklist for papers and reviewers.

11.1 Separate Pool, Context, and Reader Claims

A paper introducing a new multi-hop RAG system should explicitly distinguish three claims: a claim about the retrieved pool L_N, a claim about the budgeted context E_K, and a claim about the reader. Reporting only end-to-end accuracy can hide compensating effects. Concretely, we recommend at least one number from each of the following: pool-level support or path recall; context-level chain preservation under the budget actually used; and reader correctness under both natural and oracle or near-oracle contexts.

Table 12. Suggested minimum reporting for multi-hop RAG papers.

Reported quantity	Stage probed	Why it matters
Pool support/path recall at Top-N	Observability ($\mathcal{P}$)	Bounds the system; without it, downstream gains are unattributable.
Selected support recall at K	Selection ($\mathcal{S}$)	Reveals whether the budget preserves the chain.
Position of support in $\pi(E_K)$	Exposure ($\mathcal{O}$)	Detects lost-in-the-middle effects.
Oracle-evidence EM/F1	Fusion ($\mathcal{U}$)	Isolates reader capability from retrieval.
Deletion or counterfactual sensitivity	Faithfulness ($\mathcal{T}$)	Distinguishes causal use from parametric leakage.
Token, call, and latency cost	Efficiency / budget	Required for fair comparison across architectures.

11.2 Report Diagnostics on Conditional Subsets

Diagnostics should respect the nested-event structure of equation 28. Faithfulness numbers should be reported on the chain-present subset; exposure ablations should hold membership fixed; fusion ablations should hold both membership and ordering fixed. Numbers that mix multiple conditions hide the channel of improvement and reduce reproducibility.

11.3 Always Report Budget and Cost

Agentic and iterative methods can improve accuracy by spending more tokens, more retrieval calls, or more verifier calls. Without budget-normalized numbers, comparison is unfair. A minimal cost report includes: total tokens per question (prompt and generation), retrieval calls per question, verifier calls per question, and end-to-end latency. Where possible, accuracy-cost curves should replace single operating points.

11.4 Distinguish Gold-Used and Oracle Support

Gold supporting facts in benchmarks are not necessarily the facts a system used. Two systems with the same answer can use different chains; one chain may be valid and the other may exploit a shortcut. Reporting whether the system's actual evidence contains the gold supports, together with how often it differs, helps distinguish faithful and shortcut behavior.

11.5 Minimum Reporting Table

11.6 Threats to the Latent-Chain View

For completeness, we list four limitations of our framing. First, $C^\star$ is set-valued, not unique; some questions admit many valid chains, and partial supervision systematically undercounts them. Second, latent-reasoning methods (Section 3.8) push chain inference into model internals, where external diagnostics do not apply without additional probes. Third, agentic systems may exhibit emergent behavior that is not well captured by any fixed factorization. Fourth, our success-factor decomposition (equation 28) is exact only as a chain-rule expansion over a particular nested event sequence; under parametric leakage and shortcut events, the relationship to end-to-end accuracy is one of upper bound rather than equality. We have flagged this in the relevant theorem and remarks, and recommend that empirical work treat the decomposition as a diagnostic tool rather than as a strict prediction.

12 Conclusion

Multi-hop retrieval-augmented reasoning is at a transition point. Architectural work has matured: graph reasoning, iterative retrieval, decomposition, fusion-in-decoder, chain-of-thought, agentic search, and process supervision are well-developed, and new combinations appear at every conference. What is missing is not new architecture but a more diagnostic statistical layer that explains why systems fail and what each component is responsible for.

This survey has offered such a layer. We treat the support chain as a latent object; the retrieved pool, the selected context, the ordering, and the answer as noisy observations; and the system's job as inference under missingness, noise, budget, exposure, fusion error, and faithfulness constraints. Under this view, five statistical quantities become useful primary coordinates: chain observability, conditional evidence utility, exposure probability, fusion reliability, and causal faithfulness. Methods, benchmarks, and applications can be reorganized around these coordinates without abandoning architectural taxonomies, and the chain-rule decomposition of end-to-end success in equation 28, together with the minimax bound in Theorem 1, explains why no single module dominates and why credible empirical claims must specify which factor improved.

Three practical implications follow. First, evaluation should be conditional. Faithfulness on chain-absent examples conflates retrieval and attribution; fusion on chain-missing examples conflates selection and reasoning. Second, reporting should separate the five estimands, even when only one is the target of a paper, because joint metrics conceal compensating effects. Third, the most promising research directions are those that move multiple estimands at once without sacrificing any of them: process-supervised retrieval that improves both observability and faithfulness; bridge-aware selectors that improve both utility and exposure; and counterfactual training that improves both fusion and faithfulness.

We close by noting that the latent-chain view also clarifies what is *not* solved by the latest agentic, long-context, or latent reasoning advances. Agentic search moves the approximation from a fixed retriever to a learned policy, but it does not remove the latent-chain problem; it merely reformulates it as policy learning over evidence-acquisition actions. Long-context readers change the granularity of evidence units, but they do not eliminate exposure or faithfulness bottlenecks. Latent-reasoning methods relocate fusion into hidden states, but they introduce new auditability problems. The underlying object, the unobserved support chain, persists across these architectures. A statistical view that names it explicitly is, we believe, the simplest organizing principle for the next generation of multi-hop RAG research.

A Screening Protocol Details

Search corpus and time window. We executed structured queries over the ACL Anthology, arXiv (cs.CL, cs.IR, cs.AI, cs.LG), OpenReview, NeurIPS / ICML / ICLR / ACL / EMNLP / NAACL / SIGIR / KDD / WWW / CIKM / AAAI proceedings, and a curated list of survey venues including ACM Computing Surveys, TACL, JAIR, and TMLR. The time window covers January 2017 through January 2026, with priority emphasis on the 2022–2026 LLM-and-RAG era.

Query templates. We combined the following clusters with boolean operators: *Task* ∈ {multi-hop question answering, multi-hop reasoning, multi-hop retrieval, multi-step reasoning, compositional question answering, multi-hop fact verification, multi-step retrieval-augmented generation, agentic retrieval, multi-document reasoning}; *Mechanism* ∈ {retrieval-augmented generation, retrieve-then-read, iterative retrieval, agentic search, graph-based reasoning, knowledge graph question answering, decomposition, chain of thought, process supervision, verification, citation generation, counterfactual evaluation, long-context reading}; *Family* ∈ {dense retrieval, sparse retrieval, hybrid retrieval, reranker, planner, decomposer, reader, fusion-in-decoder, agent, tool use, reinforcement learning}.

Inclusion and exclusion. A paper was *included* if it (i) targets multi-hop tasks or any multi-hop variant such as multi-step reasoning over retrieved or graph evidence, (ii) introduces or rigorously analyzes a method, benchmark, or evaluation protocol relevant to at least one of our five estimands, and (iii) is peer-reviewed or has at least two independent third-party citations. A paper was *excluded* if its multi-hop framing is incidental, if it studies pure single-hop QA without multi-hop transfer, or if the artifact is unavailable for inspection.

Annotation. Each retained paper was annotated with (a) primary architectural family, (b) the estimands it primarily moves, (c) the evaluation regime (chain labels available, partial process supervision, end-to-end only), (d) whether faithfulness or attribution is explicitly evaluated, and (e) reported budget. This annotation underlies Sections 5 through 8 and Appendix C.

Reproducibility of the screening. We make the search queries, the inclusion/exclusion log, and the annotation schema available in the supplementary material accompanying this survey. The protocol is intended to allow future surveys to update the corpus by re-executing the same queries on later time windows.

B Mathematical Notes

Granularity of evidence units

In Definition 1, evidence units e_i can be passages, sentences, table rows, image regions, code blocks, or knowledge-graph triples. The estimands are invariant under unit refinement provided that (i) the chain length L reflects the unit granularity, and (ii) the bridge variables $b_{i,i+1}$ are defined between adjacent units at the same granularity. Mixed-granularity chains (e.g., a sentence linked to a table row via a shared entity) are permitted; the bridge variable then ranges over a heterogeneous space that includes entities, dates, numeric matches, and image-text grounding links.

Bridge priors and observability

Chain observability $\mathcal{P}(q)$ marginalizes over chain orderings. Under the often-reasonable assumption that the underlying chain is a directed acyclic graph rather than a strict sequence, observability is properly redefined as

$$\mathcal{P}(q) = \Pr\big(\{e_1, \dots, e_L\} \subseteq \mathcal{P} \mid q\big),$$

and selection utility, exposure, fusion, and faithfulness factor analogously. We keep the sequential form in the main text because all benchmarks we surveyed annotate ordered or unordered chains rather than richer DAG structures.

Decomposition under chain ambiguity

When multiple distinct chains support the same answer, the success decomposition in equation 28 should be read as a lower bound: success is at least the probability that *some* sufficient chain survives all five stages. This is the form used by oracle-style evaluators that count any complete supporting chain as correct. The minimax bound in Theorem 1 continues to apply, because the bottleneck factor governs the dominant pathway.

Identifiability under partial supervision

Proposition 1 states that $\mathcal{U}$ is identifiable conditional on $(\mathcal{P}, \mathcal{S}, \mathcal{O}) = (1, 1, 1)$. Under partial supervision, where only a subset of bridge variables is annotated, identifiability weakens but does not collapse: $\mathcal{U}$ is identifiable up to permutation on annotated bridges and remains a lower bound

Table 13. Method-to-estimand mapping (selected, non-exhaustive).

Method family	$\mathcal{P}$	$\mathcal{S}$	$\mathcal{O}$	$\mathcal{U}$	$\mathcal{T}$
Sparse retrieval (BM25, SPLADE)	$\bullet$	$\circ$			
Dense retrieval (DPR, Contriever, BGE)	$\bullet$	$\circ$			
Hybrid retrieval (HyDE, query expansion)	$\bullet$	$\circ$			
Iterative retrieval (MDR, Baleen, IRCoT)	$\bullet$	$\bullet$			
Adaptive retrieval (FLARE, DRAGIN)	$\bullet$	$\bullet$	$\circ$		
Graph traversal (GRAFT-Net, PullNet, ToG)	$\bullet$	$\bullet$	$\circ$		
GraphRAG / HippoRAG / LightRAG	$\bullet$	$\bullet$	$\circ$		
Long-context readers (LongRAG, RAPTOR)			$\bullet$	$\circ$	
Cross-encoder rerankers	$\circ$	$\bullet$			
LLM-as-reranker (RankGPT, RankZephyr)	$\circ$	$\bullet$			
Bridge-aware selectors (HopRetriever)		$\bullet$	$\circ$		
Compression (LLMLingua, RECOMP, xRAG)		$\circ$	$\bullet$		
Decomposition (DecompRC, Self-Ask, IRCoT)	$\circ$	$\bullet$		$\circ$	
Fusion-in-decoder, FiD-light				$\bullet$	
CoT/ ToT / GoT reasoning				$\bullet$	$\circ$
Latent reasoning (Quiet-STaR, Coconut)				$\bullet$	
Self-verification, CoVe, CRITIC				$\circ$	$\bullet$
PRM, RAG-Gym, MATH-SHEPHERD		$\circ$		$\bullet$	$\bullet$
Counterfactual evaluation, RARR					$\bullet$
Citation generation (WebGPT, ALCE)					$\bullet$
Agentic RAG (Search-R1, DeepRAG)	$\bullet$	$\bullet$	$\bullet$	$\circ$	$\circ$

on true fusion reliability on unannotated bridges, by an argument analogous to that of Yang et al. (2024) for hop-resolved attention attribution.

Log-scale decomposition and additive risk

Taking logs of equation 28 yields

$$\log \Pr(\text{success} \mid q) \leq \log \mathcal{P} + \log \mathcal{S} + \log \mathcal{O} + \log \mathcal{U} + \log \mathcal{T}.$$

This additive form motivates the additive-risk view: each module contributes an independent shortfall $-\log(\cdot)$ to the total deficit, and the largest term identifies the bottleneck factor for q. Practitioners can estimate each shortfall on diagnostic subsets and rank modules by remediation impact.

C Method Catalogue

Table 13 cross-references major method families discussed in Sections 5 and 6 against the five estimands they primarily move. The mapping is intentionally non-exclusive: many methods touch several estimands; we record the *strongest* effect with $\bullet$ and *secondary* effects with $\circ$. Empty cells indicate no first-order effect.

D Writing and Reporting Checklist

We summarize the reporting guidelines of Section 11 as a checklist that authors of new multi-hop RAG methods can apply before submission.

- **Estimand identification.** State which of the five estimands the proposed method primarily targets, and which it might inadvertently affect.
- **Conditional metrics.** Report retrieval coverage on chain-present examples, reader accuracy on chain-present examples, faithfulness on chain-present-and-attended examples, and corresponding chain-absent baselines.
- **Budget.** Report retrieved-token budget, prompt-token budget, total LLM calls, and wall-clock latency per query.
- **Used vs. gold support.** Report which gold passages were present in the context and which were attended to, distinguished from the full pool.
- **Process annotations.** If the benchmark provides path or process labels, report path-level metrics in addition to end-to-end metrics.
- **Counterfactual probes.** If feasible, report sensitivity to deletion or substitution of bridge evidence.
- **Failure analysis.** Provide a small table breaking down errors by latent-chain failure mode ($\mathcal{P}$-loss, $\mathcal{S}$-loss, $\mathcal{O}$-loss, $\mathcal{U}$-loss, $\mathcal{T}$-loss).
- **Threats.** State the threats to validity of the latent-chain framing for the chosen benchmarks (Section 11).

E Extended Mapping: Methods to Estimands and Datasets

Table 14 provides a finer-grained mapping for selected recent methods, listing which datasets they were primarily evaluated on, which estimands they targeted, and which estimands they inadvertently affected ('spillover'). Spillovers are particularly important for fair comparison: a retrieval-quality improvement that also reduces exposure pressure should not be credited solely as a fusion improvement.

F Claim Templates

For each estimand we provide a template that authors can use to phrase empirical claims unambiguously. The templates are intentionally verbose; their purpose is to expose the conditioning assumptions that remain implicit in the common "*method X improves multi-hop accuracy by y points*" phrasing.

Observability claim. "On queries whose annotated chain is contained in pool $\mathcal{P}$ with probability at least α (under retrieval depth k), our method increases this probability from α to α'; we do not claim improvement on queries whose chain is unobservable at depth k."

Selection-utility claim. "Conditional on chain observability ($\mathcal{P} = 1$), our method increases the probability that all chain elements appear in the reader's context window from β to β'; the increment is reported under fixed token budget B and reported reranker class."

Exposure claim. "Conditional on selection ($\mathcal{S} = 1$), our method increases the probability that all chain elements are attended to during reading from γ to γ', measured by attention-weight masking experiments at threshold τ."

Fusion claim. "Conditional on observability, selection, and exposure ($\mathcal{P} = \mathcal{S} = \mathcal{O} = 1$), our method increases the probability of producing the correct answer from δ to δ'. The remaining failures are attributed to fusion under the assumptions of Section 3."

Table 14. Extended method-to-estimand mapping with spillovers.

Method	Family	Primary	Spillover	Benchmarks
DPR	dense ret.	$\mathcal{P}$	$\mathcal{S}$	NQ, HotpotQA
Contriever	dense ret.	$\mathcal{P}$	$\mathcal{S}$	BEIR, HotpotQA
MDR	iterative	$\mathcal{P}, \mathcal{S}$	$\mathcal{O}$	HotpotQA, MuSiQue
Baleen	iterative	$\mathcal{P}, \mathcal{S}$	$\mathcal{O}$	HoVer, FEVEROUS
IRCoT	decomp.+ret.	$\mathcal{S}, \mathcal{U}$	$\mathcal{P}$	HotpotQA, 2WikiMQA
FLARE	adaptive	$\mathcal{S}$	$\mathcal{U}$	2WikiMQA, ASQA
GraphRAG	graph-RAG	$\mathcal{P}, \mathcal{S}$	$\mathcal{U}$	private, MuSiQue
HippoRAG	graph-RAG	$\mathcal{P}, \mathcal{S}$	$\mathcal{U}$	MuSiQue, 2WikiMQA
LongRAG	long-context	$\mathcal{O}$	$\mathcal{S}$	HotpotQA, MuSiQue
RAPTOR	long-context	$\mathcal{O}, \mathcal{U}$	$\mathcal{S}$	QuALITY, NarrativeQA
HopRetriever	bridge-aware	$\mathcal{S}$	$\mathcal{P}$	HotpotQA
LLMLingua	compression	$\mathcal{O}$	$\mathcal{S}$	MuSiQue, HotpotQA
RECOMP	compression	$\mathcal{O}$	$\mathcal{U}$	HotpotQA, NQ
RankGPT	reranker	$\mathcal{S}$	$\mathcal{P}$	BEIR, multi-hop
Self-RAG	adaptive	$\mathcal{S}, \mathcal{U}$	$\mathcal{T}$	ASQA, multi-hop
Self-Ask	decomp.	$\mathcal{S}, \mathcal{U}$	$\mathcal{P}$	2WikiMQA
RAG-Gym	PRM	$\mathcal{U}, \mathcal{T}$	$\mathcal{S}$	MuSiQue, HotpotQA
RARR	post-hoc	$\mathcal{T}$	$\mathcal{S}$	ASQA
CoVe	self-verify.	$\mathcal{U}, \mathcal{T}$	–	ASQA, multi-hop
ALCE	citation	$\mathcal{T}$	–	ASQA, ELI5
Search-R1	agentic-RL	$\mathcal{P}, \mathcal{S}, \mathcal{O}$	$\mathcal{U}$	HotpotQA, MuSiQue
R1-Searcher	agentic-RL	$\mathcal{P}, \mathcal{S}, \mathcal{U}$	$\mathcal{O}$	HotpotQA, MuSiQue
DeepRAG	agentic-RL	$\mathcal{P}, \mathcal{S}, \mathcal{U}$	$\mathcal{O}, \mathcal{T}$	MuSiQue, 2WikiMQA
PAR-RAG	planning	$\mathcal{S}, \mathcal{U}$	$\mathcal{P}$	HotpotQA, MuSiQue
VERITAS	verifier	$\mathcal{T}$	$\mathcal{U}$	ASQA, multi-hop
ReARTeR	re-arch.	$\mathcal{S}, \mathcal{U}$	$\mathcal{T}$	multi-hop

Faithfulness claim. "Conditional on a correct answer being produced, our method increases the probability that the answer is causally supported by the cited evidence chain from ϵ to ϵ'; causal support is measured by counterfactual deletion sensitivity at perturbation budget ρ."

Each template specifies (i) the conditioning event, (ii) the population of queries, (iii) the budget and protocol, and (iv) the auxiliary assumptions. We argue that adoption of this style would substantially reduce comparability problems across the multi-hop RAG literature.

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